

DEDICATED PRELIM PROGRAM(DPP)**UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022****DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO****TEST BOOKLET SERIES****Test Code:****DPP****TEST BOOKLET
GENERAL STUDIES
TEST -09****E9****Time Allowed: Two Hours****Maximum Marks : 200****INSTRUCTIONS**

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BYA CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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TOPICS COVERD: Wind Circulation Water in the Atmosphere, World Climate and climate change, India — Location, Structure and Physiography, Drainage System, Climate

Q.1) Consider the following statements:

1. The eye of a tropical cyclone is a region of calm and subsiding warm air.
2. Tropical cyclones generally do not have lightning.

Which of the above statement(s) is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.2) Consider the following statements about Western Disturbances:

1. Western Disturbances move towards northwest India under the influence of Tropical Easterly Jet Stream.
2. They usually get moisture from Mediterranean and Atlantic Ocean.
3. High pressure over Ukraine helps in formation of Western Disturbances.

Which of the above statement(s) is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 3 only

Q.3) Which of the following statements with reference to Omega Block is correct?

- a) It is a low-pressure pattern that blocks and diverts jet streams.
- b) Because of the Omega block, cold air from southern Europe and Africa can be pulled north.
- c) Formation of Omega block leads to formation of Spanish plume.
- d) None of the above

Q.4) Consider the following statements about effects of El-Nino:

1. It results in dry conditions and forest fires in Australia and Indonesia.
2. Fewer hurricanes over the Atlantic Ocean, the Caribbean Sea and the Gulf of Mexico.
3. Bleaching of corals in the Lakshadweep islands.

Which of the above statements are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.5) Consider the following statements about walker circulation:

1. Walker circulation is formed due to development of pressure gradient from west to east in the equatorial Pacific Ocean.
2. Years of walker circulation are beneficial for the people of Peru.
3. La-Nina is a strengthened phase of walker circulation.

Which of the above statements are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.6) Consider the following statements:

- 1) Equator divides the Llanos and Campos grasslands of Savanna.
- 2) Westerlies are dominant in shaping the climate of Savanna regions.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.7) Consider the following events:

- 1) Sunspot cycle
- 2) Volcanism
- 3) Aerosol Radiative Forcing
- 4) Milankovitch oscillations
- 5) Glacial Lake sediments

Which of the above are the astronomical causes for climate change?

- a) 1, 2, and 5 only
- b) 1, 2, and 5 only
- c) 1, 3, and 5 only
- d) 1, 2, and 4 only

Q.8) Consider the following statements:

- 1) Rice cultivation accounts for maximum Greenhouse gases emissions from the agricultural sector.
- 2) Areas with monsoonal climate largely depend upon agriculture for livelihood.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.9) Consider the following statements:

1) Deserts in South America are formed only because of Continentality.

2) Western boundaries of most African deserts are flanked by cold ocean currents.

3) Desert climates are favourable around the subtropical high-pressure belts.

Select the correct answer using the codes given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2, and 3

Q.10) Consider the following statements:

1) Crop growth is faster in winters in the Mediterranean regions due to winter rains.

2) Balearic Islands are situated at the center of the Mediterranean Sea.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.11) Consider the following statements:

1) The temperate grasslands of South America are devoid of marine influence.

2) Steppes in the Northern Hemisphere experience mild winters due to continentality.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.12) Consider the following statements:

1) Valley of flowers national park forms the core area of the Nanda Devi Biosphere reserve.

2) Glacial retreats are an important cause of the Glacial Lake Outburst floods.

3) The National Mission for sustaining Himalayan Ecosystem is under the Ministry of Environment, Forests, and Climate Change.

Select the correct answer using the codes given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.13) Consider the following pairs:

Grasslands	Country
1) Llanos	Ecuador
2) Pampas	Uruguay
3) Pustaz	Hungary
4) Canterbury	Britain

Select the correct answer using the codes given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 4 only
- d) 1 and 3 only

Q.14) Consider the following statements:

1) Temperate Monsoonal climates are mostly found in China.

2) Gulf type climate is most prominent along the coast of Yemen.

3) The grasslands in Savanna are distinctive due to the absence of trees.

Select the correct answer using the codes given below:

- a) 1 only
- b) 3 only
- c) 1 and 2 only
- d) 2 and 3 only

Q.15) Consider the following statements:

1) The whole of Tundra region is characterized by permanent snow cover.

2) MOSAiC project is designed for the study of the Antarctica climate.

Select the answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.16) Consider the following statements:

1) Coniferous trees are totally absent below the Equator due to the narrowness of continents.

2) Pine, Fir, Spruce are the major trees found in the Tundra regions.

3) Laurentian climate finds presence in the Northern Hemisphere only.

Select the incorrect answer using the codes given below:

- a) 1 and 3 only
- b) 3 only
- c) 1 and 2 only
- d) 1, 2, and 3

Q.17) Consider the following pairs:

- | | |
|---------------------------|----------|
| 1) Land of big games | Savanna |
| 2) Granaries of the world | Prairies |
| 3) Cold pole of the Earth | Siberia |

Which of the above pair(s) is/are correctly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.18) Consider the following statements:

- 1) Arctic Council is a non-governmental organization aimed at environmental protection.
- 2) Himadri is India's research base deployed in Norway for the study of the region.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.19) Consider the following statements:

- 1) Greenhouse gases are capable of absorbing short-wave radiations of the sun.
- 2) Both the Kyoto Protocol and Paris Agreement impose legally binding obligations.
- 3) India aims to have 70% of non-fossil fuel-based electricity by 2030.

Select the correct answer using the codes given below:

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) 3 only

Q.20) Consider the following:

- 1) Neem coating of urea
- 2) Soil Health card
- 3) Organic fertilizers

Which of the above is/are useful in reducing the emission of nitrous oxide(N_2O)?

- a) 1 and 2 only
- b) 1, 2, and 3
- c) 3 only
- d) 2 and 3 only

Q.21) This region is far ahead of many other areas on account of their unprecedented industrial advancement. Cyclonic disturbances ensure adequate

rainfall to these areas throughout the year. The temperature lies between 5 to 15 degree C throughout the year. The deciduous trees dominate the region and shed their leaves mainly in autumn. Which of the following climatic regions is being referred to above?

- a) Mediterranean climate
- b) Tropical Monsoon climate
- c) British Climate
- d) Steppe's grasslands

Q.22) Consider the following statements:

- 1) Steppes are found in the areas around the Black Sea and the Caspian Sea.
- 2) Thermal expansion of the ocean water is the main contributing factor to the rise in sea level.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.23) Consider the following statements:

- 1) Lake Priyadarshini is built by India around Maitri research base, Antarctica.
- 2) Polar Vortex and Polar Stratospheric clouds are directly responsible for Ozone depletion.
- 3) Kigali Amendment to the Montreal Protocol is legally binding in nature.

Select the correct answer using the codes given below:

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 only
- d) 1,2 and 3

Q.24) Consider the following statements:

- 1) The Indian Standard Meridian (ISD) cuts across the centre of Andhra Pradesh.
- 2) The smallest part of ISD falls in the state of Madhya Pradesh.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.25) Consider the following statements:

- 1) Molassis basin is an important feature in the state of Mizoram.
- 2) Mahabharat range is located between the Indo-Nepal border.
- 3) Aksai Chin forms a part of the Karakoram range.
- 4) Nubra valley is accessed through the Khardung La pass.

Select the correct answer using the codes given below:

- a) 1 and 2 only
- b) 2 and 4 only
- c) 1 and 3 only
- d) 1 and 4 only

Q.26) Which of the following statements is most appropriate regarding passes in India?

- a) The Lampiya dura pass is located to the south of Lipulekh pass.
- b) Mana pass is in the Khangchendzonga biosphere reserve.
- c) Palghat pass is situated between the Nilgiris and the Cardamom Hills.
- d) Jawahar Tunnel is also known as the Banihal pass.

Q.27) Consider the following statements:

- 1) Bhabar areas are more suitable for crop cultivation than Terai.
- 2) Khadar sediments are deposited near the flood plains.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.28) Consider the following statements:

- 1) Bhabar, Tarai, Khadar, and Bhangar together constitute alluvial plains.
- 2) Most of the Brahmaputra plains lie in the state of Arunachal Pradesh.

Select the correct answer from the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.29) Consider the following statements:

- 1) The elevation of western ghats increases southwards.
- 2) Eastern ghats are continuous with lower elevation.

- 3) Bundelkhand forms the upper part of the Deccan Plateau.

Select the correct answer from the codes given below:

- a) 1 and 2 only
- b) 1 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.30) Consider the following statements:

- 1) All the rivers of the Central Highlands originate in the Vindhya and Satpura ranges.
- 2) Malda fault exists between the Malwa Plateau and the Shillong Plateau.

Select the incorrect answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.31) Consider the following ranges/mountains:

- 1) Harishchandra
- 2) Nallamala
- 3) Dandakaranya
- 4) Ramgarh
- 5) Malya giri
- 6) Garhjat

Arrange the above ranges in the south-north direction:

- a) 2-3-1-6-5-4
- b) 2-1-3-4-6-5
- c) 2-1-3-5-6-4
- d) 1-2-4-3-6-5

Q.32) Consider the following statements regarding the Indian desert:

- 1) Absence of marine deposits indicates that the region was a part of the Peninsular plateau.
- 2) This area shows opposing slopes in the Northern and the Southern parts.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.33) Consider the following statements regarding the Indian coasts:

- 1) Natural harbors and ports are a characteristic feature of the emergent coasts.

2) Gujarat has the maximum coastline amongst all the States/UTs of India.

3) Northern Circars are the region lying between Mahanadi and Krishna

Select the correct answer using the codes given below:

- a) 1 and 2 only
- b) 3 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.34) Which of the following statements is most appropriate regarding the Islands of India?

a) The entire Island of Lakshadweep lies below the 12-degree channel.

b) Duncan Passage is situated between Swaraj dweep and the Shaheed dweep.

c) Agatti Island is situated to the north of the capital Kavaratti.

d) The highest peak of Andaman, Saddle Peak is located in the South Andaman.

Q.35) Consider the following statements regarding Andaman and Nicobar Islands:

1) These islands are an extension of the Arakan Yoma range.

2) Narcondam Island in the Andaman is a dormant volcano.

3) Mt. Thullier is the highest peak of the Nicobar group of Islands.

Select the correct answer using the codes given below:

- a) 1 and 3 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.36) Consider the following statements regarding Lakshadweep:

1) Adroit is the largest group of Islands in Lakshadweep.

2) Minicoy Island is separated from the rest of the island through a 9-degree channel.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.37) Consider the following statements:

1) Catchment area of a river is located near the source of the river.

2) Drainage basin of a river is formed by the mainstream excluding the tributaries.

3) Drainage basin confined to a large geographical area forms the watershed.

Select the incorrect answer using the codes given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1 only

Q.38) Consider the following rivers:

1) Sindh

2) Son

3) Mayurakshi

4) Ken

Arrange the rivers above in the west to east direction:

- a) 3-1-4-2
- b) 1-4-2-3
- c) 3-4-1-2
- d) 1-2-4-3

Q.39) Consider the following statements:

1) Antecedent streams maintain their course despite the upliftment of the surrounding surface.

2) Consequent streams flow according to the slope of the adjoining surface.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.40) Consider the following pairs:

Rivers	State of Origin
1) Mahanadi	Odisha
2) Kaveri	Kerala
3) Tapi	Gujarat
4) Krishna	Karnataka

Which of the above pairs is/are correctly matched?

- a) 1, 2 and 4 only
- b) 1, 3 and 4 only
- c) 3 and 4 only
- d) None of the above

Q.41) Consider the following statements:

1) River Godavari drains five states of India.

2) River Mandovi is known for delta formation.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.42) Consider the following statements:

- 1) Vamsadhara is a west flowing river forming an estuary.
- 2) Andhra Pradesh, Karnataka, Kerala forms the drainage basin of Pennar river.

Select the incorrect answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.43) Consider the following statements:

- 1) Periyar is the largest river of Kerala.
- 2) Vembanad lake is an example of lagoon lake.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.44) Consider the following statements:

- 1) The discharge of the Himalayan rivers is constant throughout the year.
- 2) Harike wetland is situated at the confluence of Satluj and Beas.
- 3) Verinag spring in Himachal Pradesh is the origin of Jhelum.

Select the correct answer using the codes given below:

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.45) Consider the following statements:

- 1) Renuka Lake is an artificial freshwater lake in Himachal Pradesh.
- 2) It is the only Ramsar wetland site of the state.

Select the correct answer using the codes given below:

- a) 1 only

b) 2 only

c) Both 1 and 2

d) Neither 1 nor 2

Q.46) Consider the following rivers:

- 1) Raidak
- 2) Rind
- 3) Sengar
- 4) Sindh
- 5) Son

Which of the above is/are the tributaries of Yamuna?

- a) 1, 2, and 4
- b) 2, 3 and 4
- c) 3, 4, and 5
- d) 1, 4, and 5

Q.47) Consider the following statements:

- 1) Rivers originating from the Amarkantak range are an example of centripetal drainage.
- 2) Rivers Narmada and Son flow in different directions despite the same source of origin.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.48) Consider the following south Indian rivers:

- 1) Penganga
- 2) Penneru
- 3) Tungabhadra
- 4) Manjira

The correct arrangement of these rivers in the North-South direction is:

- a) 1-3-4-2
- b) 1-4-3-2
- c) 4-1-3-2
- d) 4-1-2-3

Q.49) Consider the following statements:

- a) Majority of Indian rivers flow through a rift valley.
- b) Lacustrine deposits have been found from the Shivalik.
- c) There is a sharp tilt of the Indian peninsula from west to east.
- d) Most of Tapi's drainage basin lies in the state of Gujarat.

Q.50) With reference to Western disturbances, consider the following statements:

1. These are weak tropical cyclones that originate over the Mediterranean sea.
2. Jetstreams play an important role in driving these high pressure systems into India.
3. The rainfall that occurs due to these disturbances is known to be beneficial for the Rabi crops.

Which of the above statements is/are incorrect ?

- a) 1 only
- b) 2 only
- c) 1 and 2 only
- d) 1,2 and 3

Q.51) Which of the following statements is *not* correct with reference to Cloudbursts ?

- a) The Specific location and time of a cloudburst can be predicted only after the genesis of a thunderstorm has already begun.
- b) These systems have a shorter life span whereas large scale systems like tropical cyclones have a longer life span.
- c) A cloudburst can occur anytime and at any place which is affected by convective weather systems.
- d) The India Meteorological Department labels rainfall over 200 mm per hour as cloudburst.

Q.52) With reference to cyclones in India, consider the following statements:

1. Historically in India, more cyclones have impacted the Eastern coast in comparison to the Western Coast.
2. Historically in India, more cyclones have occurred in the post monsoon season in comparison to the Pre-monsoon Season.
3. All cyclones that hit the Indian coast originate in the Indian Ocean.

Which of the above statements is/are correct ?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2 and 3

Q.53) With reference to Madden Julian oscillations (MJO), consider the following statements:

1. Unlike a standing pattern like the El Niño, the Madden-Julian oscillation is a traveling pattern that propagates eastward above the warm parts of the Indian and Pacific oceans.
2. A MJO cycle can last anywhere between 120 to 150 days.

3. A MJO can completely nullify the impact of El-Nino on monsoon rains in India.

Which of the above statements is/are correct ?

- a) 1 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) 2 and 3 only

Q.54) With reference to heatwaves in India, Which of the following statements is *not* correct?

- a) In General, heatwaves are defined with reference to the normal maximum temperature of a particular region
- b) Most heat waves in India are formed over the Peninsular region especially on the leeward side of the Western ghats
- c) Inflow of Western disturbances in Northern India can influence the occurrence of heat waves in Central India
- d) A cyclone hitting the coast of Odisha can trigger a heatwave in Punjab

Q.55) According to the Indian Meteorological Department, in which of the following states/union territories does the South West Monsoon arrive first?

- a) Kerala
- b) Lakshadweep
- c) Andaman and Nicobar Islands
- d) Tripura

Q.56) Consider the following statements:

1. Norwesters are violent storms that generally occur just before sunset or just a few hours after it.
2. In India, these storms originate over Chotanagpur Plateau only.
3. In India, the rainfall caused by Norwesters is deemed harmful for the tea cultivated in Assam and the jute cultivated in West Bengal.

Which of the above statements is/are correct ?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1 and 3 only

Q.57) In India, The Western Ghats receive rain from which of the following sources on an annual basis?

1. Bay of Bengal Branch of South West Monsoon
2. Cyclones and depressions that originate in the Bay of Bengal
3. Cyclones and depressions that originate in the Arabian Sea

4. Western Disturbances

5. North East Monsoons

Select the correct answer using the code given below:

- a) 1,3 and 5 only
- b) 2,3 and 5 only
- c) 2,3 and 4 only
- d) 1,2,4 and 5 only

Q.58) With reference to the Andaman and Nicobar Islands, which of the following statements is/are incorrect?

- 1. The Andaman Islands are the extension of the Arkan yoma mountain ranges of Myanmar.
- 2. Andaman Islands are separated from Nicobar Islands by 9-degree channel
- 3. Saddle peak is the highest in Great Nicobar and Mt.Thuillier is the highest peak in North Andaman
- 4. Ross island is the largest island in the union territory

Select the correct answer using the codes given below:

- a) Only 1
- b) 1 and 2 only
- c) 2,3, and 4 only
- d) 1,2,3, and 4

Q.59) Consider the following pairs:

River	Tributary
1. Son	South Koel
2. Brahmaputra	Sankosh
3. Krishna	Musi
4. Mahanadi	Kangsabati

Select the correct pair/pairs using the code given below:

- a) 1 only
- b) 2 only
- c) 2 and 3 only
- d) 2, 3 and 4 only

Q.60) With reference to the Chambal river, Consider the following statements.

- 1. It originates along the Son river in Amarkantak Plateau of Madhya Pradesh.
- 2. Chambai is the chief tributary of the Ganga river which flows South to North direction.
- 3. Banas is the only right bank tributary of the Chambal river.
- 4. Chambal is famous for badland topography called Chambal Ravines.

Which of the statements given above is/are NOT correct?

- a) 1, 2 and 3 only

b) 2, 3 and 4 only

c) 1, 3 and 4 only

d) 1, 2, 3 and 4

Q.61) Consider the following pairs :

River	Origin
1. Alaknanda -	Satopanth Glacier
2. Ghaghara -	Mapchachungo Glacier
3. Teesta	Kangto Glacier
4. Sarda	Cho Lamu Glacier
5. Brahmaputra	Chemayungdung Glacier

Which of the pairs given above are correct?

- a) 1 and 5 only
- b) 2 and 3 only
- c) 1, 2 and 5 only
- d) 1, 2, 3 and 5 only

Q.62) Which of the following rivers do not form a radial drainage pattern?

- 1. Johilla River
- 2. Son River
- 3. Machhu River
- 4. Vamsadhara River
- 5. Luni river

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 2, 3 and 5 only
- c) 1, 3 and 4 only
- d) 4 and 5 only

Q.63) Consider the following statements.

- 1. Tamil Nadu Coast (South-East) receives heavy rains from the Bay of Bengal Branch of Indian Monsoon.
- 2. North-West region of India including Punjab, Haryana receives winter rainfall due to Tropical Easterly Jet Stream.

Which of the statements given above is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.64) Consider the following statements:

- 1. Karnataka receives more rainfall than Maharashtra from the Arabian sea branch of Indian South West Monsoon.
- 2. Chotanagpur Plateau receives miniscule rainfall from the Bay of Bengal branch of Indian South West Monsoon.

3. Tamil Nadu coast is situated parallel to the Bay of Bengal Branch of South-west monsoon.

Which of the statements given above is/are NOT correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 only
- d) 1 and 3 only

Q.65) With reference to the Indian Monsoon, Consider the following statements

1. North East Monsoon winds blow during the months of October to December and are confined to the north-east region of India.
2. North- East Monsoon Winds characterized by Oppressive heat and humidity whereas Retreating monsoon winds creates pleasant seasons with low temperature and low humidity.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.66) In the context of Indian Climate, Consider the following pairs :

Local Storms	Characteristics
1. Loo	Cold storm in the Northern plains
2. kalbaisakhi	Useful for Rice Cultivation
3. Mango showers	Pre-monsoon showers in Kerala.
4. Andhis	Evening thunderstorm in the coastal area.

Which of the pairs given above is/are correctly matched?

- a) 1 and 2
- b) 2 and 3
- c) 1 and 4
- d) 3 and 4

Q.67) Consider the following statements:

1. Torres Strait is a connecting strait between Australia and Island Tasmania.
2. Bass Strait connects Australia and Asia.

Which of the statements given above is/are correct?

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2.

Q.68) Arrange the following coasts from west to east:

1. Delta of Benin
2. Slave coast
3. Grain coast
4. Ivory coast
5. Gold coast

Choose the correct option from below:

- a) 3-4-5-2-1
- b) 1-2-3-4-5
- c) 2-3-4-5-1
- d) 4-3-2-5-1

Q.69) consider the following pairs:

Region in news	country
1. Jarablus	- Syria
2. Tigray	- Iraq
3. Najaf	- Ethiopia
4. Mandalay	- Myanmar

Which of the above pairs is/are correctly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 4 only
- d) 2 and 4 only

Q.70) Consider the following pairs:

Tropical Cyclones	origin
1. Nivar	- Bay of bengal
2. Vongfong	- Arabian Sea
3. Iota	- Eastern pacific ocean
4. Gati	- Arabian Sea

Which of the pairs given above is/are correctly matched?

- a) 1 and 3 only
- b) 1 and 4 only
- c) 2 and 3 only
- d) 2 and 4 only

Q.71) With reference to the Attorney General (AG) of India, consider the following statements:

1. According to the constitution of India, the AG is appointed by the president by a warrant under his hand and seal.
2. According to the constitution of India, the AG shall receive such remuneration as the President may determine.
3. According to the The Contempt of Courts Act, 1971, the AG's consent is required for initiation of all contempt proceedings.

Which of the above statements is/are correct?

- a) 1 and 2 only
- b) 2 only

- c) 1 and 3 only
- d) 1,2 and 3

Q.72) Which of the following adopted the “Additional Protocol to the Convention on Cybercrime” which requires action against racist and xenophobic material through computer systems?

- a) The African Union
- b) The Council of Europe
- c) The United Nations General Assembly
- d) The European union

Q.73) The term “Compulsory Licensing”, is sometimes seen in the news in the context of

- a) Electricity Distribution
- b) Data Localization Norms
- c) E-Commerce Regulations
- d) Drug Manufacturing

Q.74) The “One Health” approach recognizes that the health of people is closely connected to the health of animals. In this regard, which of the following are zoonotic diseases?

- 1. Salmonella infection
- 2. Ebola
- 3. Distemper Disease
- 4. Lyme disease
- 5. West Nile Virus Disease

Choose the correct Ans using the code given below:

- a) 1,2 and 3 only
- b) 1,2,4 and 5 only
- c) 3,4 and 5 only
- d) 1,2,3 and 5 only

Q.75) With which of the following Natural disasters would you associate the term ‘Cone of Uncertainty’, sometimes seen in news?

- a) Earthquakes
- b) Cyclones and Hurricanes
- c) Tsunamis
- d) Glacial lake Outbursts

Q.76) With reference to the Socially and Educationally Backward Classes (SEBC) in India, Consider the following statements:

- 1. According to the constitution, the Centre may or may not consult the National Commission for Backward Classes on all policy matters concerning the SEBC.
- 2. In India, the States do not have any power to identify SEBC classes.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None of the above

Q.77) With reference to the Purchasing Managers Index (PMI), consider the following statements:

- 1. It is an index of the prevailing direction of economic trends in the manufacturing sector only.
- 2. It is a perception-based index that asks the respondents about changes in some key business variables from the preceding year.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None of the above

Q.78) In the context of Indian economy, the Government has introduced the “Maximum Admissible Export Quota (MAEQ)” scheme to help deal with the surplus stocks of

- a) Onions
- b) Sugar
- c) Cotton
- d) Rice

Q.79) The principle of non-refoulement guarantees that no one should be returned to a country where they would face inhuman or degrading treatment. In this context, the right to Non-Refoulement in India is guaranteed under

- a) Article 14
- b) Article 19
- c) Article 20
- d) Article 21

Q.80) Consider the following:

- 1. United Nations Refugee Convention of 1951
- 2. Universal Declaration of Human Rights of 1948
- 3. International Covenant on Civil and Political Rights of 1966.

Which of the above conventions have been ratified by India?

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) 1 and 3 only

Q.81) India recently adopted a 'Roadmap 2030' and launched an 'Enhanced Trade Partnership' with

- Australia
- Japan
- United Kingdom
- Maldives

Q.82) With reference to Special Leave Petition, consider the following statements:

- The Indian Constitution allows the Supreme Court to grant special leave to appeal against any judgment or order in any matter or case, made by any court or tribunal in the country.
- The Special Leave Petition does not apply to any judgment or order handed down by any court or tribunal involving the armed forces.

Which of the above statements is/are correct?

- 1 only
- 2 only
- Both 1 and 2
- None of the above

Q.83) "An oxygen concentrator takes in air and separates the oxygen and delivers it into a person via a nasal cannula. Air is 79% nitrogen and 21% oxygen and a concentrator that works by plugging into a source of electricity delivers air that is upto 95% oxygen."

In this regard, which of the following materials helps in separating the Oxygen from the Nitrogen in a commercial oxygen concentrator?

- Zeolite
- Andalusite
- Kyanite
- Sillimanite

Q.84) According to the Contempt of Courts Act 1971, which of the following are *not* grounds for contempt of Court ?

- If someone Disobeys Court orders wilfully
- If someone Interferes with judicial proceedings
- If Someone tends to obstruct the administration of justice
- If Someone tends to scandalise the authority of the court

Select the correct answer using the code given below:

- 2 only
- 3 and 4 only
- 1,2,3 and 4
- None of the above

Q.85) Which of the following groups of animals enjoys the same level of protection as the Indian Pangolin in India under the Wildlife protection Act, 1972?

- Himalayn rat, Black Flying Fox, Wild Pig and Hog Deer
- Indian Tiger, Sambar deer, Indian Star Tortoise and Common Mongoose
- Mouse Deer, Chinese Pangolin, Slender Loris and Himalayn Tahr
- King Cobra, Indian Python, Nilgiri Marten and Jungle Cat

Q.86) Which of the following statements is correct regarding the constitutional office of the Governor?

- The Constitution explicitly prohibits a resident of a particular state from becoming the governor of the state.
- When requested by the Cabinet, a Governor cannot refuse to summon the Legislature under any circumstances.
- The Governor's use of discretionary powers to summon the legislature can be challenged in a court of a law.
- The Constitution has laid down elaborate grounds for removal of a governor.

Q.87) In the context of India's biodiversity, 'The Andaman Drongo', 'The Andaman Nightjar' and 'the Great Nicobar Crane', sometimes seen in news, are

- Mammals
- Birds
- Reptiles
- Crustaceans

Q.88) Consider the following statements:

- Tornadoes are columns of violently rotating air and occur in all continents of the world.
- The diameter of a cyclone is generally far greater than that of tornadoes.
- All cyclones in the world originate over the oceans and the seas.

Which of the above statements is/are correct?

- 2 only
- 2 and 3 only
- 1,2 and 3
- None of the above

Q.89) With reference to the Central Adoption Resource Authority (CARA), consider the following statements:

1. It is designated as the Central Authority to deal with inter-country adoptions in accordance with the provisions of the Prague Convention on Inter-country Adoption, 1993.
2. Under the Juvenile Justice Act, 2015, CARA primarily deals with adoption of orphan and abandoned children but not with surrendered children.

Which of the above statements is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None of the above

Q.90) Which of the following countries lies on the same longitude as the Gaza Strip ?

- a) Turkey
- b) Yemen
- c) Somalia
- d) Iran

Q.91) According to the RBI Guidelines, A 'wilful default' would be deemed to have occurred in which of the following cases?

1. A company defaults in its repayment obligations even when it has the capacity to repay.
2. A company disposes off the collateral used for the purpose of securing a term loan with the lender's knowledge.
3. A company has diverted the funds from the lender for other purposes.

Select the correct Ans using the code given below:

- a) 1 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1,2 and 3

Q.92) With reference to State Legislative councils in India, Consider the following statements:

1. Bengal was the first state where a Legislative Council started functioning in accordance with the Govt. of India Act, 1935.
2. The Constitution has capped the membership of the state legislative council to one-third of the popularly elected Legislative assembly.
3. According to the Constitution, the setting up of a state legislative council is exclusively in the hands of the state legislative assembly.
4. Unlike the Rajya Sabha, candidates having practical Experience in the Co-operative movement can be nominated to a state legislative council.

Which of the above statements are correct?

- a) 1 and 2 only
- b) 1,3 and 4 only
- c) 3 and 4 only
- d) 1,2 and 4 only

Q.93) With reference to creation of new districts in India, which of the following statements is *not* correct?

- a) The power to create new districts rests entirely with the State governments and the Central government has no role to play here.
- b) A new district can be created either through an executive order or by passing a law in the State Assembly.
- c) The power to abolish existing districts does not rest entirely with the State governments and Central Government has a role to play here.
- d) None of the Above

Q.94) The World Economic Situation and Prospects report is published by

- a) World Economic Forum
- b) United Nations Department of Economic and Social Affairs (UN DESA)
- c) International Monetary Fund
- d) World Bank

Q.95) With reference to the Central Bureau of Investigation (CBI), consider the following statements:

1. The CBI operates under the jurisdiction of the Ministry of Personnel, Public Grievances and Pensions.
2. The procedure for appointment of the CBI director is governed by the Central Vigilance Commission Act, 2003
3. Till recently, the CBI was a non-statutory body, but now it has been given statutory status under the Lokpal and Lokayukta Act, 2013.

Which of the above statements is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) 2 and 3 only

Q.96) Which of the following Central Armed Police Forces does *not* come under the Operational control of the Ministry of Home Affairs?

- a) Central Reserve Police Force
- b) Border Security Force

- c) Shahastra Seema Bal
- d) Assam Rifles

- b) 2 only
- c) 2 and 3
- d) 1,2 and 3

Q.97) Recently, Corticosteroids are in news for their application in treating COVID-19 affected patients. In this context, consider the following statements:

1. Steroid medicines are man-made versions of natural hormones and are designed to reduce inflammation.
2. Steroids increase the activity of the immune system by affecting the way white blood cells work.
3. All patients who use steroids to counter infections invariably develop side effects.

Which of the above statements is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) None of the above

Q.98) Consider the following excerpt from the RBI's recently released "Report on Management of Foreign Exchange Reserves" :

" This type of risk involves the risk of not being able to sell an instrument when required without facing significant costs. To minimize this risk, the investment strategy required here is dealing in instruments which could be converted into cash at short notice."

The type of risk Explained in the above excerpt is

- a) Credit Risk
- b) Liquidity Risk
- c) Interest Rate Risk
- d) Currency Risk

Q.99) The 17+1 Cooperation forum, sometimes seen in the news, is an investment forum consisting of 17 Central & Eastern European Countries and

- a) USA
- b) Russia
- c) China
- d) India

Q.100) With respect to river Cauvery, Consider the following statements:

1. The source of River Tungabhadra lies to the south of the source of River Cauvery.
2. Bhavani, Kabini and Hemavathi rivers are tributaries of river Kaveri.
3. River Cauvery eventually flows into the Gulf of Mannar.

Which of the above statements is/are correct?

- a) 1 only

Q.1) Ans: C

Exp:

- **Statement 1 is correct:** The eye of a tropical cyclone is a region of calm and subsiding warm air. There is little or no precipitation and sometimes blue sky or stars can be seen.
- **Statement 2 is correct:** Ordinary thunderstorms develop vertically, meaning upright from the ground. In the tropical cyclones, there is high rotational energy and air does not climb up directly. Instead, it takes a swirly, roundabout path. So, while they reach the height of 10 to 12 kilometers the rising motion isn't quite as strong to produce lightning.

Reference:

<https://www.sciencenewsforstudents.org/article/explainer-what-is-eyewall-of-hurricane-or-typhoon>

Q.2) Ans: C

Exp:

- **Statement 1 is incorrect:** Western Disturbances move Eastward towards northwest India and Pakistan under the influence of Subtropical Westerly Jet Stream and Westerlies.
- **Statement 2 is correct:** The moisture in Western Disturbances usually comes from Mediterranean Sea and Atlantic Ocean. On their way they pick up moisture from Caspian Sea and Black Sea as well.
- **Statement 3 is correct:** High pressure over Ukraine and neighbourhood areas causes the intrusion of cold air from the Polar region towards Mediterranean Sea. This generates favourable conditions for cyclogenesis in the upper atmosphere, which promotes the formation of an eastward-moving extra tropical depression.

Reference:

<http://data.conferenceworld.in/IEEERAJSHREE/P772-777.pdf>

Q.3) Ans: C

Exp:

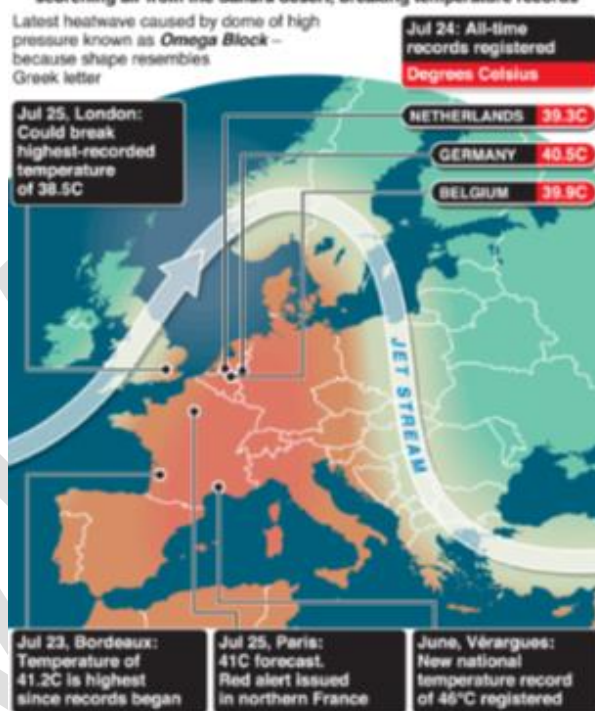
- **Option A is incorrect:** Omega block is a high-pressure pattern that blocks and diverts jet streams.
- **Option B is incorrect:** When an omega shaped wave is present on the jet stream which arcs over Europe, warm dry air from southern Europe and

Africa can be pulled north, pushing temperatures higher than normal.

- **Option C is correct:** An important consequence of this phenomena is formation of Spanish plume, in which very warm air moving northwards from Spain towards the UK rises and leads to formation of cumulonimbus clouds and heavy rainfall.

Heatwave smashes European records

For the second time in a month, a high pressure system is drawing scorching air from the Sahara desert, breaking temperature records



Reference:

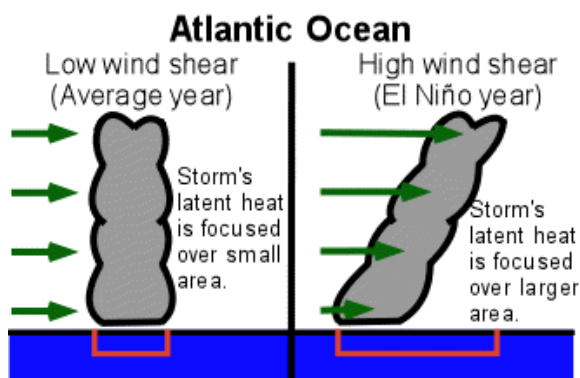
OnlyIAS July 2019 current affairs

<https://www.metoffice.gov.uk/weather/learn-about/weather/types-of-weather/storms/spanish-plume>

Q.4) Ans: D

Exp:

- **Statement 1 is correct:** El-Nino results in floods in Peru while Pacific coast of Australia and Indonesia have abnormally dry conditions that forces the collapse of Agriculture with instances of Forest fire.
- **Statement 2 is correct:** During an El-Nino year, fewer hurricanes form over the Atlantic Ocean, the Caribbean Sea and the Gulf of Mexico due to increased wind shear. The increased wind shear helps to prevent tropical disturbances from developing into hurricanes.



- **Statement 3 is correct:** During El-Niño year sea surface temperature of western Indian Ocean increases and this leads to bleaching of corals. Such large scale bleaching was seen in 1998 and recently as well.

Reference:

<https://www.firstpost.com/tech/science/lakshadweep-corals-are-dying-and-theyll-take-the-islands-inhabitants-with-them-5402301.html>
[http://ww2010.atmos.uiuc.edu/\(Gh\)/guides/mtr/hurr/enso.xml#:~:text=The%20primary%20Exp:%20for%20the,over%20the%20Caribbean%20and%20Atlantic.](http://ww2010.atmos.uiuc.edu/(Gh)/guides/mtr/hurr/enso.xml#:~:text=The%20primary%20Exp:%20for%20the,over%20the%20Caribbean%20and%20Atlantic.)

Q.5) Ans: B

Exp:

- **Statement 1 is incorrect:** The walker circulation is a Convective cell of air circulation which is formed due to the development of a pressure gradient from East to West in the equatorial Pacific Ocean.
- **Statement 2 is correct:** Walker circulation facilitates Upwelling of cold water near Peru coast resulting in High Pressure atmospheric stability and dry weather. Also upwelling of nutrients from below the ocean facilitates fishing in this region, which is beneficial for the people of Peru. Otherwise, during the El-Niño period, due to break in upwelling of water, fishes die due to want of nutrients or they migrate.
- **Statement 3 is correct:** La-Niña is an opposite phenomenon of El-Niño and is a strengthened phase of walker circulation.

Reference:

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Q.6) Ans: A

Exp:

- **Statement 1 is correct:** The Savanna are the tropical grasslands characterized by a distinct wet and dry season. They are most developed in Sudan and hence also called the Sudan climate. They are found in West African Sudan, Eastern Africa, and the southern parts of Africa above the tropic of Capricorn. There are two distinct regions of Savanna both North and South of the equator in South America. *Llanos* are found near Orinoco basin whereas *Campos* are found in the Brazilian highlands.
- **Statement 2 is incorrect:** The dominant winds in the Savanna regions are the **Trade winds** which bring rainfall to the coastal districts of these areas.

Q.7) Ans: D

Exp:

- **Statement 1 is correct:** Sunspot cycles play an important role in shaping the climate of the Earth. They appear as dark and cooler patches on the surface of the Earth. The increase in sunspots on the sun is associated with cooler and wetter climatic conditions while their decrease is associated with warmer and drier conditions of climate on earth.
- **Statement 2 is correct:** Volcanic eruptions are responsible for the release of excessive hot gases, and aerosols on the earth's atmosphere. Increase in such eruptions causes greater accumulation of these materials resulting in warming of the Earth.
- **Statement 3 is incorrect:** Aerosol Radiative Forcing is a method of quantifying the effect that aerosol produces on the climate. It is found that excessive emission of aerosol by human activities is responsible for changing the earth's climate. Hence, it is not a natural rather an anthropogenic factor.
- **Statement 4 is correct:** Milankovitch oscillations are associated with the changes in the axial tilt, revolution of the earth around the sun, eccentricity. These factors together are responsible for changing the amount of insolation received on Earth and hence influence the climate of the Earth.
- **Statement 5 is incorrect:** The sediments found in the glacial lakes are not directly responsible

for the change of Earth's climate rather they give information regarding the alternate dry and wet climates experienced by the Earth.

Q.8) Ans: B

Exp:

- **Statement 1 is incorrect:** Agriculture sector is largely responsible for the emission of greenhouse gases. The highest percentage of greenhouse gases from agriculture is added by the Livestock products. Rice accounts for the second largest share in the emission of Greenhouse gases.
- **Statement 2 is correct:** Agriculture is the prime occupation of the people living in the areas of Monsoonal climates. Crops like rice, sugarcane, jute, etc. are grown in these areas.

Q.9) Ans: B

Exp:

- **Statement 1 is incorrect:** The driest desert in South America is the **Atacama Desert** or the Peruvian desert located in Chile and Peru and flanked by the cold Peruvian current. The **Patagonian Desert** is located in Argentina, Chile, and the Falkland Islands and is due to the rain shadow area formed by being on the leeward side of the Andes. Hence, all the deserts in South America are not formed due to continentality.
- **Statement 2 is correct:** Western boundaries of the African deserts like the Sahara and the Namib Deserts is flanked by the cold water currents. **Canaries current** near the Sahara desert and the **Benguela current** near the Namib flank the boundaries of these deserts.
- **Statement 3 is correct:** Hot deserts are the hottest regions of the world. They are mostly found on the western coasts of the continents between 15 to 30 degrees North and South of the equator. They lie near the **Horse latitudes** or the **Sub-tropical high-pressure belts** which experience descending air and negligible precipitation. The diurnal range of temperature is extremely high in the deserts.

Q.10) Ans: D

Exp:

- **Statement 1 is incorrect:** The Mediterranean regions are characterized by the warm, dry summers and cool, moist winters. The rainfall in the Mediterranean regions occurs in winter months. Despite the rainfall, the growth is slow during the winters. The climate of the Mediterranean regions supports the growth of a wide variety of crops.
- **Statement 2 is incorrect:** **Balearic Islands** belonging to Spain are located above Algeria in the Mediterranean Sea. They are located in the Western part of the Mediterranean Sea and not at the Center.

Q.11) Ans: D

Exp:

- **Statement 1 is incorrect:** Temperate grasslands are called **Steppes** in Eurasia. They are also found in North and South America. In North America, they are called as **Prairies** while in South America they are called as **Pampas**. These grasslands are mostly found in the interior of the continents and are away from the maritime influence of the sea. However, Pampas of South America in Argentina and Uruguay experience the maritime influence of the sea.
- **Statement 2 is incorrect:** The steppes in the Northern Hemisphere are mainly under the influence of continentality. The summers are very warm while the winters are very cold due to this. However, in the Southern Hemisphere, the winters are mild.

Q.12) Ans: A

Exp:

- **Statement 1 is correct:** **Nanda Devi Biosphere Reserve** in the state of Uttarakhand is a World Heritage Site under the UNESCO's Man and Biosphere Programme. Its core areas are Nanda Devi and the Valley of Flowers National Park.

- **Statement 2 is correct:** **Glacial Lake Outburst floods** are caused due to the sudden release of water from the accumulated glacial lakes triggered by an avalanche, earthquake, landslides, or any other anthropogenic reasons. Glacial retreats are responsible for the creation of an impression on the earth's surface which gets filled with debris and water forming a lake. This lake on crossing its threshold causes flooding. Hence, glacial retreats are an important cause of Glacial Lake Outburst Floods.
- **Statement 3 is incorrect:** National Mission for Sustaining Himalayan Ecosystem is a sub mission under the National Action Plan on Climate Change. It is under the **Ministry of Science and Technology**.

- **Statement 3 is incorrect:** The grasslands of Steppes are characterized by the absence of trees with shorter grasses whereas the grasslands of Savanna are characterized by tall grasses with short trees.

Q.15) Ans: D

Exp:

- **Statement 1 is incorrect:** The permanent snow cover in the Polar or the Arctic climate is restricted to the higher reaches of Greenland. The lowlands on the other hand have few months which are ice free and are characterized by Tundra. These regions are located in Alaska, coastal areas of Greenland, Northern Canada, and the Arctic areas of Eurasia.
- **Statement 2 is incorrect:** **MOSAIC or the Multidisciplinary Drifting Observatory for the Study of Arctic Climate (MOSAIC)** led by the **Alfred Wegener Institute** in Germany and designed by the **International Arctic Science Committee (IASC)**. It is a yearlong expedition in the Central Arctic region that aims to study the climate of the Arctic.

Q.13) Ans: B

Exp:

- **Statement 1 is incorrect:** Llanos are the tropical grasslands in South America on the Orinoco basin. It is associated with **Columbia and Venezuela**.
- **Statement 2 is correct:** Pampas are the temperate grasslands of South America associated with **Argentina and Uruguay**.
- **Statement 3 is correct:** Pustaz are the temperate grasslands situated in **Hungary**.
- **Statement 4 is incorrect:** Canterbury grasslands are the temperate grasslands in **New Zealand**.

Q.16) Ans: B

Exp:

- **Statement 1 is incorrect:** Narrowness of the continents in the Southern Hemisphere are responsible for reducing the severity of the winters due to the maritime influence. Regions with Siberian climates are present in the Cool Temperate Continental areas and are absent in the Southern Hemisphere. Coniferous trees can withstand such cold temperatures and are hence found in the Siberian regions. However, in the Southern Hemisphere such trees are not totally absent but can be seen in the uplands of South Chile, New Zealand, Tasmania, southeast Australia.
- **Statement 2 is incorrect:** Pine, Fir, Spruce are the common coniferous forests found in the Siberian areas. Tundra regions comprise the lowest form of vegetation in the form of lichens, mosses, etc.
- **Statement 3 is correct:** Laurentian climates are the cool temperate eastern margins of the continents

Q.14) Ans: A

Exp:

- **Statement 1 is correct:** The Warm Temperate Eastern Margins are also called the China type of climate. It is found on the Eastern margins of the continents in the temperate regions. The summers are moist and warm, and the winters are cool and dry with maritime influence. There is evenly distributed rainfall throughout the year.
- **Statement 2 is incorrect:** The Gulf climate is found along the temperate eastern margins in South-east USA along the borders of the Gulf of Mexico and hence is also called as the Gulf Climate.

in the Northern Hemisphere. They are found only in certain areas of North America and Asia (near China, Korea, Siberia, north Japan). They are absent totally in the southern hemisphere.

Q.17) Ans: D

Exp:

- **Statement 1 is correct:** The African Savanna regions are a home to a number of wild animals, which are trapped and hunted by tourists every year. Hence, it is also called the land of big games.
- **Statement 2 is correct:** Steppes are known as the granaries of the world as these grasslands provide for the cultivation of wheat, maize, etc. and follow extensive mechanized cultivation.
- **Statement 3 is correct:** The Siberian climate is experienced in the Northern Hemisphere and is absent in the Southern Hemisphere. They are defined by long cold winters and short cool summers. They are also called the cold pole of the world.

Q.18) Ans: B

Exp:

- **Statement 1 is incorrect:** Arctic Council is an **intergovernmental** organization that works to promote Environmental protection and Sustainable development of the Arctic region. It has 8 member states: **USA, Canada, Denmark, Russia, Iceland, Norway, Sweden, and Finland**. India was given an **observer status** in the year 2013.
- **Statement 2 is correct:** India's research base, **Himadri**, for the study of the Arctic is deployed in Norway. Besides this India also has an underwater observatory, **IndARC** for studying the Arctic climate and its linkages with the monsoon.

Q.19) Ans: B

Exp:

- **Statement 1 is incorrect:** Greenhouse gases are important in contributing to global warming. Earth is heated by the insolation from the sun and from the radiation emitted from the surface of the Earth. Incoming solar radiation is shortwave radiation while the long wave radiations are emitted by the Earth's surface. Greenhouse gases like carbon dioxide, water vapor, methane, etc. absorb these short-wave radiations and cause the heating of the Earth's surface leading to Global warming.
- **Statement 2 is correct: Kyoto Protocol (2005) and Paris agreement (CoP 21, 2015)** impose legally binding obligations on the member states. They are a part of **UNFCCC** (United Nations Framework Convention on Climate change) that was signed in Rio at **Earth's Summit in 1992**.
- **Statement 3 is incorrect:** Indian NDC (National Determined Contributions) include:
 - A reduction in emission intensity by **33-35% of GDP from 2005 levels**.
 - **40%** share of non-fossil fuel based sources in the share of electricity by **2030**.
 - Additional carbon sink of **2.5 to 3 billion tonnes** of Carbon dioxide.

Q.20) Ans: A

Exp:

- **Statement 1 is correct:** Neem coating of the urea helps in reducing the emissions of nitrous oxide.
- **Statement 2 is correct:** Soil Health card monitors the health of the soil and determines the optimum amount of fertilizers required by the soil. It prevents excessive use and hence controls the emission of nitrous oxide from the overuse of nitrogenous fertilizers.
- **Statement 3 is incorrect:** The most dominant cause for nitrous oxide emission is the agricultural production which includes the use of nitrogenous fertilizers. Even organic fertilizers can contribute to emissions due to the use of manure obtained by livestock.

Q.21) Ans: C

Exp:

- **Statement C is correct:** The above description is of British type of climate. It is also referred to as the Cool Temperate Western Margins and is experienced in Britain and the low lying areas of north west Europe. The mean annual temperature ranges between 5 to 15 degree C and seldom rises beyond 18 degrees. It is characterized by high rainfall throughout the year owing to the cyclonic activity with the peak rainfall occurring during winter or autumn months. These areas are under the influence of westerlies and have warm summers. The dominant trees are the deciduous trees which shed their leaves in the winter to avoid frost and snow. They are the industrialized and advanced areas of the world.

Q.22) Ans: C

Exp:

- **Statement 1 is correct:** Steppes are the tropical grasslands found in the temperate continental areas. They are present in the areas around the Black sea and the Caspian Sea. They are called Steppes in Eurasia, Pustaz in Hungary, Prairies in North America, Downs in Australia, Velds in Africa, and Pampas in South America. They are characterized by short grass and absence of trees. They are referred to as the Granaries of the World.
- **Statement 2 is correct:** Thermal Expansion of the water is one of the main reasons for the rise in sea level as warmer water would occupy greater space. Other factors are: melting glaciers, changes in groundwater level, etc.

Q.23) Ans: D

Exp:

- **Statement 1 is correct:** Lake Priyadarshini is a freshwater lake built by India around its research station in Antarctica, Maitri.
- **Statement 2 is correct:** Polar Vortex is the cool low pressure area formed in the lower stratosphere. It is weak during summers but strongest during winters. At very low temperatures during winters, the lower temperature of the vortex leads to the formation of Polar Stratospheric clouds. These

clouds act as a substrate for the ozone depleting substances.

- **Statement 3 is correct:** Vienna Convention for the protection of the ozone layer was signed in 1985. The **Montreal Protocol** to the Vienna convention was signed in 1987. Both are the first and the only global treaty to achieve universal ratification. The **Kigali amendment** to the Montreal Protocol was signed in 2016 which is legally binding in nature.

Q.24) Ans: A

Exp:

- **statement 1 is incorrect:** The Indian Standard Meridian (ISD) passes through the states of **Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha,** and **Andhra Pradesh**. It passes an area in the eastern Andhra Pradesh.
- **Statement 2 is correct:** ISD covers least distance in the state of Madhya Pradesh. It traverses through its Eastern Boundary.

Q.25) Ans: D

Exp:

- **Statement 1 is correct:** Mizoram is also known as the Molassis basin made of soft unconsolidated deposits.
- **Statement 2 is incorrect:** Mahabharat range is located in Nepal above the Shivalik range. Hence, it does not form a boundary between India and Nepal.
- **Statement 3 is incorrect:** Aksai Chin is an Indian territory under the occupation of China, located in the UT of Ladakh. It is a part of Ladakh Plateau.
- **Statement 4 is correct:** Nubra valley is a triangular valley formed with the confluence of river Nubra and Shyok. It can be accessed through the Khardung La pass, which is located in Leh region of Ladakh.

Q.26) Ans: D

Exp:

- **Statement 1 is incorrect:** The Lipulekh pass is located at the trijunction of India-Tibet and Nepal. Lampiya dura is also located in the state of Uttarakhand but northern to the Lipulekh.
- **Statement 2 is incorrect:** Mana pass is in the Indian state of Uttarakhand in the Nandadevi Biosphere reserve. Khangchendzonga Biosphere Reserve is a mixed Heritage site under the Man and Biosphere Programme located in the state of Sikkim.
- **Statement 3 is incorrect:** Palghat pass is located in Kerala. It is located between the Nilgiris in the north and the Annamalai hills to the south.
- **Statement 4 is correct:** Banihal Pass in Jammu and Kashmir is located in the Pir Panjal range. It is also called the Jawahar tunnel. It connects the Kashmir valley to the outer Himalayas.

Q.27) Ans:

Exp: B

- **Statement 1 is incorrect:** Bhabar is a narrow area running parallel to the foothills of the Shivalik. It is located at the end of its slope and hence river deposits are associated with this area. The streams or rivers usually disappear in this area. Terai areas are found south of the Bhabar areas and the streams or rivers emerge in these areas. Hence these areas are more suitable for the cultivation of crops as compared to Bhabar.
- **Statement 2 is correct:** The alluvial plains are subdivided into Khadar and Bhangar. Bhangar are the older alluvium and are deposited away from the flood plains of the rivers. Whereas Khadar are the new alluvium deposits which are subjected to frequent flooding. Hence, they are more fertile with finer silt deposition.

Q.28) Ans: D

Exp:

- **Statement 1 is incorrect:** The northern plains are characterized by Bhabar, Terai and the alluvial plains. The alluvial plains are further divided into Khadar and Bhangar. Hence,

Bhabar and Terai do not form a part of the alluvial plains.

- **Statement 2 is incorrect:** Most of the Brahmaputra plains lie in the state of Assam and not Arunachal Pradesh.

Q.29) Ans: A

Exp:

- **Statement 1 is correct:** Western Ghats are continuous with higher elevation as compared to Eastern Ghats. The height of the western ghats increases southwards with the highest peak **Anamudi** (Kerala) located on the Annamalai hills of the Western ghats.
- **Statement 2 is incorrect:** Eastern ghats are discontinuous and lower in elevation as compared to the western ghats. It is highly eroded and eastwards flowing rivers like Mahanadi, Godavari, Krishna, Kaveri forms delta.
- **Statement 3 is incorrect:** Bundelkhand region forms a part of the Central Highlands of the Peninsular Plateau and not the Deccan Plateau. The region is associated with the states of Uttar Pradesh and Madhya Pradesh.

Q.30) Ans: C

Exp:

- **Statement 1 is incorrect:** Vindhya and Kaimur range are the source of origin of several rivers. Most of the tributaries of Yamuna originate from here. However, Banas river originates from the Aravallis and is an exception here.
- **Statement 2 is incorrect:** Malda fault exists between the Rajmahal Hills/Chotanagpur Plateau and the Meghalaya/Shillong Plateau. It is responsible for the detachment of the Northeastern Plateau from the Deccan Plateau. The northeastern plateau consists of Garo, Khasi, Jaintia hills in Meghalaya and the Karbi Anglong hills of Assam.

Q.31) Ans: C

Exp:

- **Statement C is correct:** Harishchandra range is located in the state of Maharashtra, Nallamala hills form a part of the Eastern Ghats in Andhra Pradesh, Dandakaranya is associated with Odisha, Chhattisgarh, Telangana and Andhra Pradesh, Ramgarh hills are found in the state of Jharkhand, Malyagiri hills are located in the state of Odisha, and Garhjat range are the mountain ranges stretching across Odisha, Jharkhand, and Chhattisgarh.

Q.32) Ans: B

Exp:

- **Statement 1 is incorrect:** The presence of a wood fossil park at Atkal and the marine deposits of Brahmsar, Jaisalmer indicates the region was submerged under the sea and contains marine deposits. The underlying rocks of the region appear as an extension of the peninsular plateau.
- **Statement 2 is correct:** The Northern part of the desert area is sloping towards Sindh while the southern part has a slope towards Rann of Kachchh in Gujarat. Hence, the area shows opposing slopes.

Q.33)

Ans: B

Exp:

- **Statement 1 is incorrect:** Western coasts of India are an example of submergent coasts. Natural ports and harbours are formed along the submergent coasts. Kandla, Mangalore, Cochin, Mormagao, JLN port, etc. are some of the examples of natural ports.
- **Statement 2 is incorrect:** Gujarat has the maximum coastline among the mainland states. If the Union territories are also considered, then Andaman and Nicobar Islands have the maximum coastline.
- **Statement 3 is correct:** Northern circars formed a part of the former British territory in the Madras presidency. They are found along the states of Andhra Pradesh and Odisha. It is a region lying between Mahanadi and Krishna rivers.

Q.34) Ans: C

Exp:

- **Statement 1 is incorrect:** Some parts of Lakshadweep Island lie above the 12-degree channel.
- **Statement 2 is incorrect:** Duncan passage in the Andaman is situated between south Andaman and the Little Andaman.
- **Statement 3 is correct:** Agatti Islands of Lakshadweep is situated to the north of the capital Kavaratti.
- **Statement 4 is incorrect:** Saddle peak is the highest peak of Andaman and is situated in the North Andaman.

Q.35) Ans: D

Exp:

- **Statement 1 is correct:** The Andaman and Nicobar group of Island are an extension of Arakan Yoma range which passes through Myanmar and the Indian states of Assam, Nagaland, Manipur, and Mizoram.
- **Statement 2 is correct:** Narcondam Island is situated in the North Andaman is an example of a dormant volcano.
- **Statement 3 is correct:** Mt. Thullier is the highest peak of the Nicobar group of Islands. It is located in the Great Nicobar group of Island.

Q.36) Ans: C

Exp:

- **Statement 1 is correct:** The largest island in Lakshadweep is the Adrott Island. Minicoy island is the second largest island of Lakshadweep.
- **Statement 2 is correct:** Minicoy Island is the southernmost Island of the Lakshadweep and is separated from the rest of the Lakshadweep through the 9-degree channel.

Q.37) Ans: B

Exp:

- **Statement 1 is correct:** Catchment area can be defined as the area from where the river gets its water. Hence, it is located near the source of the river.
- **Statement 2 is incorrect:** Drainage basin of a river is the entire geographical area drained by a river including its tributaries.
- **Statement 3 is incorrect:** Drainage basin of a river is composed of a larger area while the watersheds are the small areas. The catchment areas of small rivulets or rills is regarded as the watershed.

Q.38) Ans: B

Exp:

- **Statement B is correct:**
 - River **Sindh** originates from the Malwa Plateau and is a tributary of Yamuna. It flows through the states of Uttar Pradesh and Madhya Pradesh.
 - River **Son** originates from the Amarkantak plateau, Madhya Pradesh. It is the second largest tributary of Yamuna and meets Ganga at Arrah, Bihar.
 - River **Ken** is the tributary of Yamuna and flows through the Bundelkhand region of Uttar Pradesh and Madhya Pradesh.
 - **Mayurakshi** river originates in the Trikuta hills of Jharkhand. Hence, from west to east, the correct order of the above rivers is Sindh, Ken, Son, and Mayurakshi.

Q.39)

Ans: C

Exp:

- **Statement 1 is correct:** Ganga river is an example of the antecedent river. These rivers originate prior to the upliftment of the surrounding landmass and maintain their original course cutting through the mountainous region.
- **Statement 2 is correct:** Consequent streams on the other hand flow as per the slope of the

adjoining surface or mountains from where they originated.

Q.40) Ans: D

Exp:

- **Statement D is correct:**
 - The origin of river **Mahanadi** is near Sihawa, Chhattisgarh. It flows through the states of Madhya Pradesh, Chhattisgarh, and Odisha.
 - The origin of river **Kaveri** is in Brahmagiri hills, near Kodagu, Karnataka. It flows through the states of Karnataka, Kerala, and Tamil Nadu.
 - The origin of river **Tapi** is in Betul, Madhya Pradesh and flows through the states of Maharashtra, Madhya Pradesh, and Gujarat.
 - River **Krishna** originates near Mahabaleshwar, Maharashtra and passes through the states of Maharashtra, Karnataka, and Andhra Pradesh.

Q.41) Ans: A

Exp:

- **Statement 1 is correct:** River **Godavari** is the largest river in the Peninsular river system. It originates near Nashik, Maharashtra and flows through the states of Maharashtra, Madhya Pradesh, Chhattisgarh, Odisha, and Andhra Pradesh. The major tributaries of Godavari are Penganga, Manjira, Pranhita, and Indravati.
- **Statement 2 is incorrect:** River **Mandovi/Mahadayi** originates at the Bhimgad, Karnataka and flows through the states of Goa, Maharashtra, and Karnataka. It is regarded as the lifeline of Goa. It is a west flowing river that drains into Arabian Sea and does not form a delta. The famous Dudhsagar falls are located on this river.

Q.42) Ans: C

Exp:



- **Statement 1 is incorrect:** Vamsadhara is an east flowing river that originates in Odisha. It flows through the states of Odisha and Andhra Pradesh.
- **Statement 2 is incorrect:** Pennar is an east flowing river that originates in Nandi hills in Karnataka. It flows through the state of Karnataka and Andhra Pradesh.

- **Statement 1 is incorrect:** Renuka lake is the largest natural lake situated in the state of Himachal Pradesh. It is a Ramsar wetland site.
- **Statement 2 is incorrect:** Renuka wetlands are not the only wetland sites of Himachal Pradesh. There are three such sites in Himachal Pradesh, namely, **Pong Dam** in Kangra, **Renuka** in Sirmaur, and **Chandertal** in Lahaul and Spiti.

Q.43) Ans: B

Exp:

- **Statement 1 is incorrect:** Bharathapuzha is the largest river of the state of Kerala while Periyar is the second largest. Bharathapuzha originates near Annamalai hills and flows through the states of Kerala and Tamil Nadu. It drains at the Lakshadweep sea.
- **Statement 2 is correct:** Vembanad lake is the longest lake in India and the largest lake of Kerala. **Vembanad-Kol wetlands** is the second largest wetland site in India after the Sundarbans. It is a part of backwaters of Kerala, locally called as *Kayals*. This is an example of a lagoon. Lagoons are the shallow coastal water bodies separated from the sea by a series barrier islands/coral reef, etc.

Q.44) Ans: B

Exp:

- **Statement 1 is incorrect:** Himalayan rivers are perennial, but their discharge is not constant throughout the year. Ganga river experiences maximum discharge during the months of August/September which coincides with the Indian monsoons while it has minimum discharge during January to June months.
- **Statement 2 is correct:** **Harike wetlands** are located at the confluence of Satluj and Beas rivers. It is located in Punjab and is a Ramsar site.
- **Statement 3 is incorrect:** Verinag spring at the foothills of Pir Panjal range near Kashmir valley is the origin of Jhelum. Verinag spring is in Anantnag district of Jammu and Kashmir.

Q.45) Ans: D

Exp:

Q.46) Ans: B

Exp:

- **Statement B is correct:** Among the given rivers, Rind, Sengar, and Sindh are the tributaries of Yamuna. Raidak river is a tributary of Brahmaputra. Son river is the second largest tributary of Ganga after river Yamuna.

Q.47) Ans: C

Exp:

- **Statement 1 is correct:** Rivers originating in the Amarkantak Plateau are an example of the radial drainage system. **Radial** drainage system is formed when a river originating from a hill discharges its water in all the directions. **Centripetal** drainage patterns are formed when the rivers discharge their waters from all the directions to a particular lake or a depression.
- **Statement 2 is correct:** Both river Narmada and Son originate from the Amarkantak plateau. Narmada flows to the Arabian Sea while joins Ganga as its tributary, which eventually drains at the Bay of Bengal.

Q.48) Ans: B

Exp:

- **Statement B is correct:** River **Penganga** is a tributary of Godavari and has its origin in Maharashtra. **Penneru** river originates from the Nandi hills of Karnataka. River **Tungabhadra** is a tributary of Krishna. **Manjira** is a tributary of Godavari that originates near Balaghat in Ahmednagar, Maharashtra. The correct sequence

of these rivers in the north-south direction is Penganga, Manjira, Tungabhadra, and Penneru.

Q.49) Ans: B

Exp:

- **Statement 1 is incorrect:** Narmada and Tapi are the west flowing rivers that flow through the rift valley between the Vindhya and the Satpura ranges.
- **Statement 2 is correct:** Lacustrine (related to lakes) are found from the Shivalik. They are important in tracing the evolution of the Himalayan river system.
- **Statement 3 is incorrect:** There exists a slight (not sharp) tilt in the Deccan Plateau from west to east, which is responsible for most of the rivers draining the Bay of Bengal.
- **Statement 4 is incorrect:** River Tapi originates in Betul, Madhya Pradesh and flows through the states of Maharashtra, Madhya Pradesh, and Gujarat. Nearly 79 percent of its drainage basin is in the state of Maharashtra.

Q.50) Ans: C

Exp:



- **Statement 1 is not correct:** Western disturbances are **weak temperate or Extra-Tropical (not tropical) frontal cyclones** that **originate** over the Mediterranean sea, gain moisture from the Caspian Sea and the Black Sea and cause winter rainfall in the **North Western part of India**. The Himalayas act as **orographic barriers**.
- **Statement 2 is not correct:** The **subtropical westerly jet stream** drives this **low pressure** (cyclones cannot develop in High pressure

regions as these regions are characterized by subsidence of wind) system towards the east of the Mediterranean sea across Afghanistan, Pakistan and North West India. These storms are residual frontal cyclones which move at the height of 2000 metres from the mean sea level.

- **Statement 3 is correct:** Rabi crops during the **winter season** are greatly influenced by the **arrival of western disturbance**. It brings **cloudiness** and **precipitation** in many parts of north India during rabi season. Rabi crops, particularly wheat crop, down under rainfed conditions is **greatly benefited by the arrival of Western Disturbances**.

Q.51) Ans: D

Exp:

- **Statement 1 is correct:** Large scale features, which are **conductive** for occurrence of severe thunderstorms associated with cloudburst, **are predictable two to three days in advance**. However, the **specific location and time of cloudburst can be predicted in a few hours in advance**, when the **genesis of a thunderstorm has already commenced**.
 - To detect these sudden developments, a **Doppler Weather Radar (DWR)**, a powerful tool for time and location specific prediction of cloudburst, can be deployed a few hours in advance. **Coupled with satellite imagery** this can prove to be useful inputs for extrapolation of cloudbursts anywhere in India.
- **Statements 2 and 3 are correct:** A cloudburst can occur anytime and at any place which is **affected by convective weather systems**. Surrounded by oceans on three sides, **peninsular India is a favoured location** for the genesis of convective weather systems. In a short span of time, if the right combinations of atmospheric conditions like **instability, moisture content and triggering mechanisms are available**, cloudbursts are **possible**.
 - It is **however not necessary** that cloudbursts happen **only in the**

mountains or high altitudes as the weather systems have **compatibility** and **criticality** in space and time.

- This implies that the **smaller scale weather structures like tempest, tornado, cloudburst, etc. have shorter life span** whereas large scale **systems like tropical cyclones have a longer life span.**

- **Statement 4 is incorrect:** The IMD classifies rainfall intensity as **20-30 mm rain per hour** as an intense spell; **30-50 mm per hour** as a **very intense spell**; **50-100 mm per hour** as an **extremely intense spell**; and over **100 mm per hour** as a **CLOUDBURST**.

- These massive coagulated clouds with heavy water content hover over a very small location unlike storms that spread anywhere from five to 50 km range and convert into a downpour. **The dead weight of the cloud is so massive and unbearable** that it simply **collapses under its own weight** with such **ferocity** and accompanied by **thunder and lightning**.

- Monsoon withdrawal in India officially begins in early September. Usually, **soon after the withdrawal process begins**, Cyclones begin to form in either areas of **Arabian Sea** or **Bay of Bengal**.

- During the late summer months, **the sea surface temperatures reach their highest levels** and provide tropical cyclones with the **energy** they need to **develop** into major storms. **Wind shear can tear disturbances** apart before they strengthen, which gradually fades later in the season, **reaching a minimum in mid to late September**. The **lack of shear combines** with the increase in air temperature, water temperature and building moisture in the tropics.

- In contrast, **these factors do not align in an optimum manner during the pre monsoon season (April-May)** and hence the number of cyclones in these months is considerably low.

- As a matter of fact, **Cyclone Fani (2019, Odisha)** is only the **second severe cyclone in the last 118 years** to form in the **Bay of Bengal (BOB)** in the month of **April** and cross over to the Indian mainland. Between 1891-2017, **only 14 severe tropical cyclones formed in April** over BOB and only one storm crossed the Indian mainland."

- **Statement 3 is correct:**

- One of the prime reasons why the Bay of Bengal is more prone to Cyclogenesis is that the **Bay of Bengal**

Q.52) Ans: A

Exp:

- **Statement 1 is correct:**
 - Both the **Bay of Bengal and Arabian Sea** experience cyclonic events, thanks to their proximity to the Indian Ocean, where cyclones are a common phenomenon. However, when the two are compared, the **Bay of Bengal sees approximately five times as many cyclones as its Western counterpart**. In addition, **cyclones in the Bay are stronger and deadlier**. And what's more, **nearly 58% of cyclones formed in the Bay of Bengal reach the coast** as compared to **only 25% of those formed in the Arabian Sea**.

- **Statement 2 is correct:**

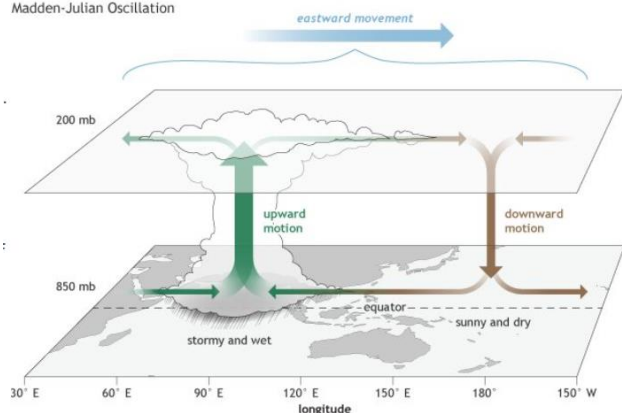
also welcomes cyclones formed far away in the Pacific Ocean. Cyclones usually weaken over a landmass. But due to lack of any such land between the Pacific Ocean and the Bay of Bengal, cyclonic winds easily move into the bay. Thus **Not all cyclones originate in the Indian ocean.**

when the MJO passes over the Indian Ocean. But the el Nino's impact is felt across India while the madden-julian oscillation's impact is limited to only some parts of India for a period of 30 to 60 days. Therefore, **a complete nullification is not possible.**

Q.53) Ans: A

Exp:

Madden-Julian Oscillation



• **Statement 1 is correct and 2 is incorrect:**

- An MJO is an oceanic-atmospheric phenomenon which affects weather activities across the globe. It brings major fluctuation in tropical weather on weekly to monthly timescales.
- The MJO can be defined as an eastward moving 'pulse' of clouds, rainfall, winds and pressure near the equator that typically recurs every 30 to 60 days. It's a traversing phenomenon and is most prominent over the Indian and Pacific Oceans.

• **Statement 3 is incorrect:**

- The Indian Ocean Dipole (IOD), El Nino and MJO are all oceanic and atmospheric phenomena, which affect weather on a large scale. The El-Nino is known to cause droughts in India. However if the periods of El Nino overlap with the convective phase of the Madden Julian oscillations, the effect of El Nino can be reduced to some extent especially in the Arabian Sea cost regions where it rains heavily

Q.54) Ans: B

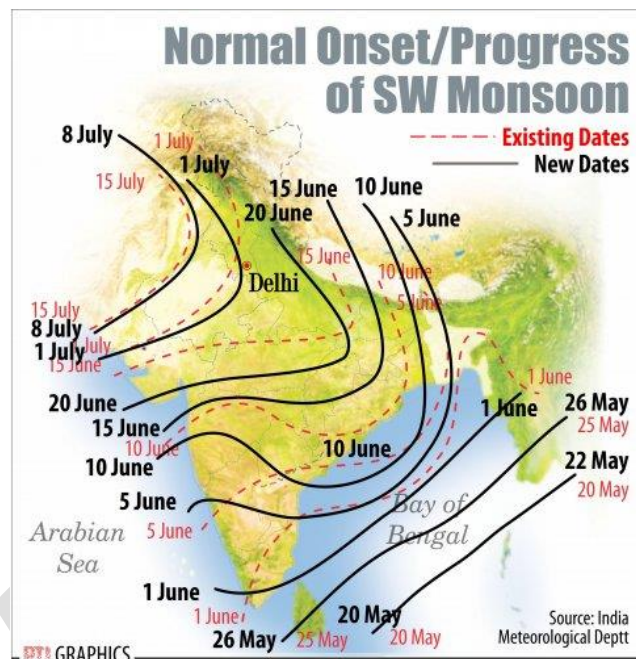
Exp:

- **Statement 1 is correct:** Heat Waves occur over India between March and June. Meteorologists define a heatwave event in reference to the normal maximum (day) temperature for a location.
 - For example, Heatwaves are declared in the plains if the temperature crosses 40 degrees Celsius. Over the hills, the threshold temperature is 30 degrees Celsius. When the day temperature jumps by 4 to 5 degrees above the normal maximum temperature of a location, it is declared as a heatwave.
- **Statement 2 is incorrect:** Heatwaves are common over the Core Heatwave Zone (CHZ) — Rajasthan, Punjab, Haryana, Chandigarh, Delhi, West Madhya Pradesh, Uttar Pradesh, Chhattisgarh, Orissa, Vidarbha in Maharashtra, parts of Gangetic West Bengal, Coastal Andhra Pradesh and Telangana, as categorised by India Meteorological Department.
 - The regions in the extreme north, northeast and southwestern India are less prone to heatwaves. Heat waves are rare over the peninsular region because of Maritime conditions.
- **Statement 3 is correct:** Summer season reaches its peak by May 15 in India, when the day temperatures across north, west, and central India cross 40 degrees and hover close to 45 degrees then on. In 2020, north India did not experience such temperatures till May 21.
 - It was mainly because of the continuous inflow of Western Disturbances that influenced the weather in the north till as late as April. There have been frequent passing of Western Disturbances over

the north, appearing after every five to seven days. Originating in the Mediterranean Sea, Western Disturbances are eastward-moving winds that blow in lower atmospheric levels. They thus affect the local weather of a region during its onward journey.

- **Statement 4 is correct:** In 2020, Cyclone Amphan, which hit the coasts of Northern Odisha and West Bengal, had a huge role in triggering heat waves across North India. Cyclone Amphan, which was a massive SuperStorm covering 700 kms, managed to drag maximum moisture from over the Bay of Bengal, entire South Peninsula, parts of Central India and to some extent, even from the Arabian Sea

- All the moisture that was otherwise built during the thunderstorm and rainfall, got gradually depleted from over vast areas as the storm advanced towards West Bengal and Bangladesh. It thus triggered dry north-westerly winds to blow over Rajasthan, Punjab, Madhya Pradesh, Uttar Pradesh and Maharashtra causing severe heatwave.



- As can be seen in the map here, the Monsoon hits the Andaman and Nicobar islands, a full week before the Arabian Sea branch hits the Kerala coast (mainland of India).
- Western Rajasthan is the last place (Mid July) where the monsoon reaches - which partially explains the arid conditions.

Q.56) Ans: A

Exp:

- **Statement 1 is correct:** Norwesters are isolated rainfall and thunderstorm events which occur in India and Bangladesh, often with violent hurricane-speed winds. Typically, such storms originate in a low-pressure area during the peak summer season. The Low pressure intensifies when the sun beats down on the region. Therefore, they generally occur just before sunset or just a few hours after it, when thick dark black clouds start appearing over the South Western sky and then bring gale-speed wind with torrential rain, often with hail, but spanning only a short period of time.
- **Statement 2 is incorrect:** Kal Baisakhi (Norwesters in the Gangetic Plains) originates over Bihar and Jharkhand area, in the chota nagpur plateau and moves eastwards and strikes West Bengal and Odisha. These are

Q.55) Ans: C
Exp:

extremely severe in nature and take Chhattisgarh under its purview as well.

- However, they are also common in Kerala, Tamil Nadu and coastal Karnataka. The Mango Showers or the CHERRY BLOSSOM SHOWERS of Kerala in the month of April and May are examples of norwesters .
- **Statement 3 is incorrect:** The rainfall in these storms is beneficial for pre-kharif crops grown in India. The tea cultivated in Assam and the jute and rice and tea cultivated in West Bengal and Bangladesh benefit from these showers. As a matter of fact, the reason they are named Mango showers or cherry blossom showers in the South Indian States is because they help and their ripening.

• **Option 2 and 3 are correct**

- The Western Ghats face the full brunt of cyclones originating in the Arabian Sea (Nisarga, 2020) and the low pressure depressions which intensify into deep depressions.
 - Interestingly, Cyclones Originating in the Bay of Bengal can also cause rain in the Western Ghats. For example cyclone Amphan in 2020, caused rains in Karnataka, Kerala and Tamil Nadu. Similarly cyclone Gaja, hit the coast of Tamil Nadu in 2018 and wreaked havoc in the districts of Coimbatore, Dindigul (Western ghats) etc.
 - Similarly, Deep depressions in Bay of Bengal can cause rainfall in Kerala and Tamil Nadu in the Western Ghats Region.

• **Option 4 is incorrect:**

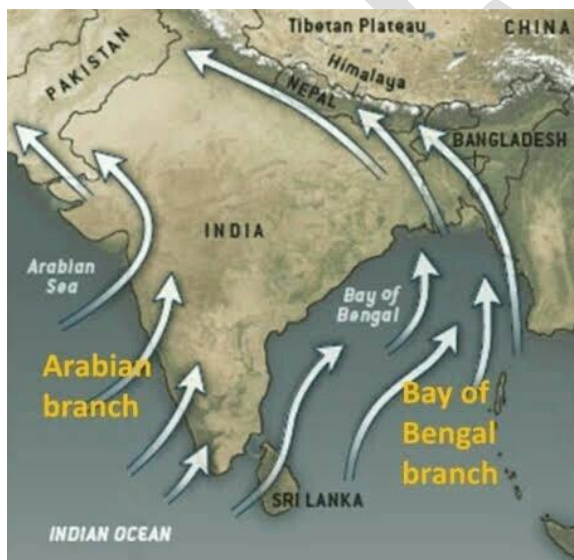
- The western disturbances are generally limited to the northern and North Western parts of the country. Hence, Western ghats do not get rainfall from these weak extra-tropical cyclones. At times however, when western disturbances become more intense in the Indian Region, they can extend even up to 15 degree north Latitude, resulting in rainfall in Western Maharashtra (Western ghats), Gujarat and the entire Madhya Pradesh to the south. However, this is not an annual occurrence and is an exception rather than a norm.

• **Option 5 is correct:**

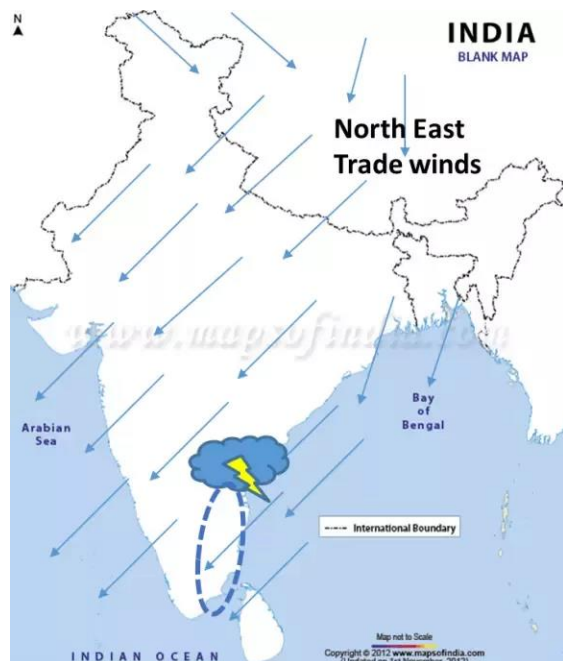
Q.57) Ans: B

Exp:

- **Option 1 is incorrect**
-



- As can be seen in the figure here, the Bay of Bengal Branch travels almost parallel to the eastern ghats and meets the Eastern Himalayas head on. From there on, it turns left and enters the Gangetic plain while making its way towards Punjab, Haryana and even Jammu and Kashmir. Thus, there is no rainfall in Western Ghats, caused by the Bay of Bengal branch.



- The **North Eastern monsoon** winds pick up moisture from the Bay of Bengal and have enough force to cause rain in the **interior districts of Tamilnadu (Nilgiris, Coimbatore, Theni) and even parts of Kerala.**

Q.58) Ans: C

Exp:

Statement 1 is correct – Andaman and Nicobar Islands were formed due to the collision between Indian Plate and Burma Minor Plate. Therefore, Andaman Islands were an extension of the submerged **Arakan Yoma** mountains of Myanmar.

Statement 2 is wrong - Andaman and Nicobar Islands are separated by a **10 degree** channel.

8 degree channel	Minicoy and Maldives
9 degree channel	Minicoy and Lakshadweep
10 degree channel	Andaman and Nicobar Islands

Statement 3 is wrong: Saddle peak is the highest in North Andaman and Mt.Thuillier is the highest peak in the Great Nicobar Islands.

Statement 4 is wrong: **Havelock island** is the largest island in the union territory. Recently it was renamed as Swaraj Dweep.

Q.59) Ans: C

Exp:

Pair 1 is Incorrect : **Son river** is a perennial river located in central India **originating in the Amarkantak hills** in Madhya Pradesh and finally merges with the Ganga in Bihar. Tributaries of Son river -**Rihand River, Johila River, North koel River, Gopad River, Kanhar river.** South Koel River, a tributary of the Brahmani River.

Pair 2 is correct : The river Brahmaputra originates in Himalayan Lake Mansarovar in Tibet to the outfall in the Bay of Bengal. Flowing southwest, it receives its main left bank tributaries, viz., Dibang or Sikang and Lohit; thereafter, it is known as the Brahmaputra. Among the tributaries Subansiri, Manas, Jia Bharali, Pagladiya, Puthimari and **Sankosh** etc. are snow fed.

Pair 3 is correct : The principal tributaries joining Krishna are the Ghataprabha, the Malaprabha, the Bhima, the Tungabhadra and Musi. Musi River is a tributary of the Krishna River in the Deccan Plateau flowing through Telangana state in India. **Hyderabad stands on the banks of Musi river.**

Pair 4 is Incorrect : The Mahanadi is one of the major east flowing peninsular rivers draining into the Bay of Bengal. The principal tributaries of the Mahanadi river are; Sheonath, Jonk, Hasdeo, Mand, Ib, Ong and Tel. **Kangsabati is a tributary of Damodar river.**

Source of reference : NCERT , Oxford Student Atlas

UPSC question referred for this: 2019 Question on Kangsabati Dam, 2016 Question on Brahmaputra Tributary

Q.60) Ans: A

Exp:



Statement 1 is Incorrect : The Chambal river originates at Janapav, south of Mhow town, on the south slope of the Vindhya Range in Madhya Pradesh. Chambal flows north-northeast through Madhya Pradesh, running for a time through Rajasthan, then forming the boundary between Rajasthan and Madhya Pradesh.

Statement 2 is Incorrect : The Chambal River moves into Uttar Pradesh and runs for around 32 km prior to meeting the Yamuna River in Etawah district of Uttar Pradesh at an altitude of 122 m, creating a portion of the greater Gangetic drainage system.

Statement 3 is Incorrect : The Banas is a river which lies entirely within the state of Rajasthan in western India. It is a left bank tributary of the Chambal River, itself a tributary of the Yamuna, which in turn merges into the Ganga. The Banas is approximately 512 kilometres in length.

Statement 4 is correct : Badland topography is a major feature of the Chambal valley is characterized by an undulating floodplain, gullies and ravines. Ravines are a type of fluvial erosional feature and are formed as a result of constant vertical erosion by streams and rivers flowing over semi-arid and arid regions.

Source of reference : NCERT

UPSC question referred for this: UPSC 2018 Question on Teesta river.

Q.61) Ans: C
Exp:

Pair 1 is correct : The Ganges as Alaknanda rises in the southern Himalayas on the Indian side of the Tibet border in the Satopanth Glacier and Bhagirath Kharak Glaciers. The source of the Alaknanda lies in the ranges above the temple town of Badrinath.

Pair 2 is correct : Ghaghara originates in the glaciers of mapchachungo glaciers. After collecting the waters of its tributaries- Tila, Seti and Beri, it comes out of the mountain, cutting a deep gorge at Shishapani. The river sarda or kali joins it in the plain before it finally meets the Gnaga at Chhapra.

Pair 3 is Incorrect : Teesta river originates at Pahunri (Teesta Kangse) Glacier in eastern Himalaya and flows through Indian state of Sikkim and West Bengal and finally joins Brahmaputra in Bangladesh to further

drain into Bay of Bengal. River Rangeet originated in Sikkim and is the main tributary of the Teesta.

Pair 4 is Incorrect : River Sarda or Saryu rises in the Milam glaciers in the Nepal Himalayas where it is known as Goriganga. Along the Indo-Nepal border, it is called kali or chauk, where it joins the Ghaghara.

Pair 5 is correct : Brahmaputra river originates in the Chemayungdung Glacier of the Kailash range near the Mansarovar Lake. It is entered by the names of Siang and Dihang in India. And after it is joined by its two main tributaries, the Dibang and the Lohit, it is known by the name of Brahmaputra. It flows in Bangladesh by the name of Jumna.

Source of reference : NCERT

UPSC question referred for this: 2019 UPSC PRELIM SET A Question number 38

Q.62) Ans: D
Exp:

Statements 1 and 2 are correct : Amarkantak Hills Radial Drainage Pattern - River Son, River Narmada, River Johilla and River Mahanadi. The Amarkantak region is a unique natural heritage area and is the meeting point of the Vindhya and the Satpura Ranges, with the Maikal Hills being the fulcrum. This is where the Narmada River, the Son River and Johila River (Tributary of Son) emerge.

Statement 3 is correct : Rajkot Hills Upland Radial Drainage Pattern - Machhu, Bhagawan, Bhandar, Shetrunji and River Mandav.

Statement 4 is incorrect : Vamsadhara River is an east-flowing river which originates in Kalahandi district of Odisha, flows in Odisha, along its boundary with Andhra Pradesh and finally joins the Bay of Bengal at Kalingapatnam, Andhra Pradesh. It is the main river of north-eastern Andhra region and the Boddepalli Rajagopala Rao Project was constructed on it to meet the irrigation needs of the region.

Statement 5 is incorrect : Luni river Originates in the Naga hills of Aravalli Ranges near Pushkar valley in Ajmer District. It is also known as Sagarmati. After crossing Govindgarh, it joins the Saraswati river which originates from Pushkar lake and from then on it is

referred to as the Luni River. The Luni river is Not part of any Radial Drainage system.

Source of reference : NCERT and INDIAN GEOGRAPHY Khullar

UPSC question referred for this: 2019 UPSC PRElim SET A Question number 38

Q.63) Ans: D

Exp:

Statement 1 is Incorrect : Tamil Nadu Coast (South-East) receives heavy rains accompanied by stormy winds in the winter sometimes causing great damages to the crops and disrupting the means of transport. Rainfall comes from the **North East Monsoon** and the Bay of Bengal Branch of Indian Monsoon as it flows parallel to the coast.

Statement 2 is Incorrect : North West region of India including Punjab, Haryana, Jammu and Kashmir , Himachal Pradesh , Northern Rajasthan and western Uttar Pradesh receive moderate to heavy rainfall due to **westerly depression or western disturbances coming from Mediterranean sea**. It is beneficial for Early ripening of Rabi crops, especially Wheat. Himalaya receives a considerable amount of snowfall in winter.

Q.64) Ans: C

Exp:

Statement 1 is correct : Karnataka's Mountain topography is broader than Maharashtra. Greater Mountain width makes Rain clouds travel a longer distance and hence more time for water to coalesce and precipitate as rainfall. Compared to this, Narrower width of Western Ghats in Maharashtra causes Rain bearing winds to cross over to the leeward side quickly before rainfall can occur.

Statement 2 is Incorrect : Arabian sea branch of Indian South west monsoon strikes the coast of Mumbai. Moving along the Narmada and Tapi valleys, these winds cause rainfall in extensive areas of central India. **The Chotanagpur Plateau receives a miniscule 15**

cm rainfall from Arabian sea branch.

Thereafter they enter Ganga plains and mingle with the Bay of Bengal Branch.

Statement 3 is correct : Tamil Nadu coast situated **parallel to the Bay of Bengal Branch** of South-west monsoon. It lies in the **rainshadow area of the Arabian sea branch** of the South west Monsoon. Rainfall comes from **North East Monsoon** and Not Bay of Bengal Branch of Indian Monsoon

Q.65) Ans: D

Exp:

Statement 1 is Incorrect : India receives rainfall during two seasons. About 75 per cent of the country's annual rainfall is received from the Southwest monsoon between June and September. North- East Monsoon occurs during October to December, and is confined to the Southern peninsula. It is also called the winter monsoon and it is important for Tamil Nadu, Puducherry, coastal Andhra Pradesh etc.

Statement 2 is Incorrect : **North- East Monsoon Winds** characterized pleasant seasons with low temperature and low humidity and clear skies. They do not give extreme rain to any part of India except Tamil Nadu Coast. On the Other hand, **Retreating South West monsoon winds are characterized** by Oppressive heat and humidity known as **"October Heat"**.

Extra Edge by Only IAS

- North East Monsoon winds blow from the North east Direction from the Land to the sea.
- Retreating South West Monsoon winds blow from the South West direction but are not strong enough to blow right into the northern plains. They withdraw in stages which results in decreasing rains.

Q.66) Ans: B

Exp:

Pair 1 is Incorrect : Loo is **hot , dry and oppressing winds** blowing in the **northern plains** from Punjab to Bihar with higher intensity between Delhi and Patna.

Pair 2 is correct : Kalbaisakhi (Calamity of the month of baisakh) are dreaded evening thunderstorms in Bengal and Assam. It is **also called Norwesters**. These showers are **useful for tea, rice and jute cultivation**. In Assam, these storms are known as **"Bardoli Cheerha"**.

Pair 3 is correct : Mango Showers are pre-monsoon showers which are common phenomena in Kerala and Karnataka. They help in early ripening of mangoes.

Pair 4 is Incorrect : The strong dust storms resulting from the convective phenomena are locally known as **andhis (blinding storms)**. They move like a **solid wall of dust and sand**. The wind velocity often reaches 50-60 kmph and the visibility is reduced to a few metres. **Such dust storms are common in Rajasthan, Haryana, Punjab, Jammu region, Delhi, Uttar Pradesh and Bihar.**

Extra Edge by Only IAS

Blossom showers- with these pre-monsoon showers, coffee flowers blossom in Kerala and nearby areas.

Q.67) Ans: D

Exp:

Statement 1 is Wrong: Torres strait is the Strait that connects Australia with Asia.

Statement 2 is wrong: Bass strait is the strait that connects Australia with Island Tasmania.



Q.68) Ans: A

Exp:

The **Gold Coast** derived its name since it was rich in gold, petroleum, sweet crude oil and natural gas, which was situated in West Africa in the Gulf of Guinea.

Ivory Coast, also known as **Côte d'Ivoire**, officially the **Republic of Côte d'Ivoire**, is a country located on the south coast of **West Africa**. The name "Ivory Coast" reflected the major trade that occurred on that particular stretch of the coast: the export of ivory.

Grain Coast, present in the west coast of Gulf of Guinea and extends till Liberia and derives its name from the spice trade.

The **Slave Coast** is a historical name for that part of coastal West Africa along the Bight of Benin. It derives its name from the Atlantic slave trade where the africans are taken into slave trade from the early 16th century to the late 19th century.

Q.69) Ans: C

Exp:

Pair 1 is correct - Jarablus is in Northern **Syria**. Recently Russia attacked this place which was in control of Turkey backed opposition forces.

Pair 2 is incorrect - Tigray is in **Ethiopia** where the local government had seized federal government facilities in November 2020 which sparked a civil war like situation. This has led to lakhs of people getting displaced and a humanitarian crisis.

Pair 3 is incorrect - Najaf is a city in central **Iraq** where **Pope Francis met Ayatollah Al-Sistani** who is the top Islamic Shiite cleric in Iraq. This region had witnessed severe violence after the rise of IS.

Pair 4 is correct - Mandalay is a historic city in **Myanmar** where **protests are ongoing** to restore democracy and against the coup by the Myanmar military.

Source: Indian express, The Hindu

UPSC Q referred: Pre 2020 Q78 set B

Q.70) Ans: B

Exp:

Option 1 is correct - Nivar impacted mainly the Tamil nadu coast with effects felt in neighbouring Andhra. It weakened before the landfall from “very severe” to “severe” category cyclone.

Option 2 is incorrect - It was a typhoon which impacted the Islands of Philippines (called Typhoon Ambo locally) and nearby regions in SouthEast and East Asia. It originated in the Western pacific.

Option 3 is incorrect - It was the latest Atlantic hurricane to attain category 5 intensity. It caused lots of damage in central America.

Option 4 is correct - It was a “very severe” category storm which was also the strongest on record to make landfall in Somalia. It emerged from the Arabian sea.

Source:

<https://www.nationalgeographic.org/encyclopedia/el-nino/>

UPSC Q referred: Pre 2020 set B Q53

Q.71) Ans: B

Exp:

- **Statement 1 is incorrect and Statement 2 is correct:** According to article 76 of the constitution, “ the president shall appoint a person who is qualified to be appointed as a judge of the supreme court to be Attorney General for India.” Hence the president is **not mandated to appoint AG by a warrant under his hand and seal** (unlike the CAG, the Governor and the Supreme Court and High Court Judge)
 - The same article stipulates that the AG shall hold office during the **pleasure of the president** and shall receive such remuneration as the President may determine.
- **Statement 3 is incorrect:** Section 15 of the The Contempt of Courts Act, 1971, lays down the procedure on how a case for contempt of court can be initiated.

- In the case of the **Supreme Court, the Attorney General or the Solicitor General**, and in the case of **High Courts, the Advocate General**, may bring in a **motion** before the court for **initiating a case of criminal contempt**.
- However, **if the motion is brought by any other person**, the **consent in writing** of the **Attorney General** or the Advocate General is required.
 - Thus, The AG’s consent is mandatory when a private citizen wants to initiate a case of contempt of court against a person
- However, **when the court itself initiates a contempt of court case**, the **AG’s consent is not required**. This is because the court is exercising its **inherent powers under the Constitution (Article 129 and Article 215)** to punish for contempt and such Constitutional powers cannot be restricted because the AG declined to grant consent.
 - The Supreme Court’s recent judgment in “**Vijay Kurle & Ors, being Suo Motu Contempt Petition (Criminal) No. 2 of 2019**” settles the position that in suo motu criminal contempt cases, the consent of **Attorney General is not required**.

NEWS : [explained: Why is consent of A-G required to initiate contempt proceedings? | explained News, The Indian express](#)
[‘An icon of India’s legal system’: Tributes pour in for Soli Sorabjee | India News, The Indian express](#)

Q.72) Ans: B

Exp:

- **Additional Protocol to the Convention on Cybercrime:**
 - Additional Protocol to the Convention on Cybercrime, concerning the criminalisation of acts of a racist and xenophobic nature committed through computer systems is an **follow**

up protocol to the **Council of Europe Convention on Cybercrime**, also called the **BUDAPEST CONVENTION on CYBERCRIMES**. Final text of this protocol was adopted by the **Council of Europe Committee of Ministers** on 7 November 2002.

- The Protocol requires participating States to criminalize the dissemination of racist and xenophobic material through computer systems, as well as of racist and xenophobic-motivated threats and insults.
- Two of the English speaking states in Europe, **Ireland and the United Kingdom**, have not signed the additional protocol.

NEWS: [On global cybercrime, India votes in favour of Russia-led resolution | India News, The Indian express](#)
+ **BLACK LIVES MATTER MOVEMENT** (always in news)

Question Style:

IAS /2019

Which of the following adopted a law on data protection and privacy for its citizens known as 'General Data Protection Regulation' in April 2016 and started implementation of it from 25th May, 2018?

- ☐ Australia
- ☐ Canada
- ☐ The European Union
- ☐ The United States of America

Q.73) Ans: D

Exp:

- **Compulsory Licensing:**
 - When the Govt. Authorities license companies or individuals **other than the patent owner to use the rights of the patent** — to make, use, sell or import a product under patent (i.e. a patented product or a product made by a patented process) — **without the permission of the patent owner** - it is called **Compulsory licensing**.
 - Allowed under the **WTO's TRIPS (intellectual property) Agreement** provided certain procedures and conditions are fulfilled.

- The Supreme Court recently asked the **Central Government** why it was not considering invoking the powers under the **Section 92 of the Patents Act, 1970 for compulsory licensing, for drugs like Remdesivir, Tocilizumab, Favipiravar**, which are used for treating COVID-19 patients.

- **Section 92 of the Patents Act** is a special provision enabling the Central Government to issue **compulsory licenses** for the manufacture of patented drugs in a **public health emergency**.
- **Section 100 of the Patents Act** enables the Central Government to use **patented inventions for government purposes**.

NEWS : [Why Centre Not Considering Compulsory Licensing For COVID Drugs Like Remdesivir, Tocilizumab? Supreme Court Asks \(livelaw.in\)](#)

Q.74) Ans: B

Exp:

- **Options 1,2,4 and 5 are correct:**

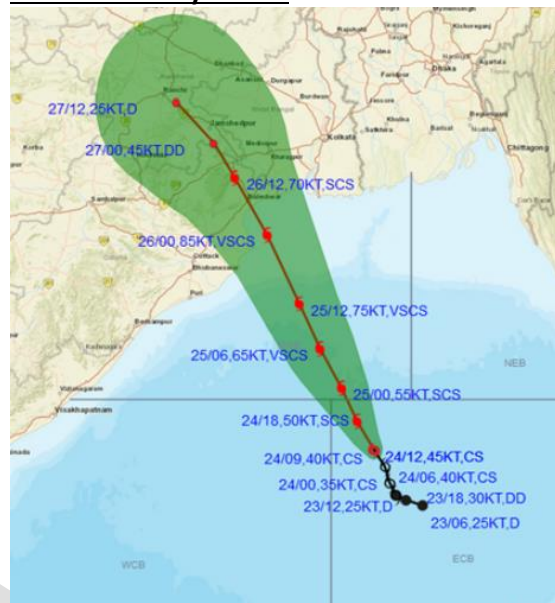
- **ONE HEALTH APPROACH**
 - One Health is an approach that recognizes that the health of people is closely connected to the health of animals and our shared environment.
 - The areas of work in which a One Health approach is particularly relevant include **food safety, the control of zoonoses** (diseases that can spread between animals and humans, such as flu, rabies and Rift Valley Fever), and combating **antibiotic resistance** (when bacteria change after being Exp:osed to antibiotics and become more difficult to treat).

- The earth has Experienced changes in climate and land use, such as deforestation and intensive farming practices. Disruptions in environmental conditions and habitats can provide new opportunities for diseases to pass to animals.
- These changes have led to the spread of existing or known (endemic) and new or emerging zoonotic diseases, which are diseases that can spread between animals and people. Examples of zoonotic diseases include:

- **Salmonella infection**
- **West Nile virus infection**
- Q Fever (Coxiella burnetii)
- Anthrax
- Brucellosis
- **Lyme disease**
- Ringworm
- **Ebola**
- Rabies
- Nipah
- Severe Acute Respiratory Syndrome
- Middle East Respiratory Syndrome

- The Distemper disease, caused by the **CANINE DISTEMPER VIRUS**, is not **zoonotic** in nature. Recently, several lions and leopards succumbed to the **CDV in the Gir Forests of Gujrat**.

• Prediction of Cyclones:



- The **cone of uncertainty** represents the **probable track of the center of the hurricane/cyclone**, based on the models used to make the forecast. The cone represents the **uncertainty in the forecast** of the **storm's center**, not necessarily the areas that will Experience impacts. The cone of uncertainty is used to get a rough idea of where the storm center may go, and the current wind watches and warnings for coastal area
- In the adjoining figure, the **area marked in green is the "cone of uncertainty"** which helps the authorities in taking preparedness and evacuation measures much before the cyclone actually makes landfall and ravages the coastal regions.

NEWS: [Cyclone Tauktae, Yaas: How cyclone prediction techniques work, why they are important and how to prepare for storms - The Financial express](#)

Q.76) Ans: B

Exp:

- **Statement 1 is incorrect:** According to **Article 338B (102nd Constitution Amendment Act in 2018)**, The Union and every State Government shall consult the **National Commission for Backward Classes** on all major policy matters

NEWS: [A 'One Health' approach that targets people, animals - The Hindu](#)

Q.75) Ans: B

Exp:

affecting the socially and educationally backward classes.

- **Statement 2 is correct:** Recently, In the judgment that declared the Maratha reservation unconstitutional, a Constitution Bench of the Supreme Court, by a **3:2 majority**, ruled that after the passage of the **102nd Constitution Amendment Act** in 2018, the **States do not have any power to identify** 'socially and educationally backward' (SEBC) classes.
 - The Court interpreted the bare text of the Amendment to the effect that **only the President can publish a list of backward classes** in relation to each State and **that only Parliament can make inclusions or exclusions** in it (Article 342A).

NEWS: [explained | The Supreme Court ruling on identifying backward classes - The Hindu](#)

Q.77) Ans: D

Exp:

- **Statement 1 is incorrect:** PMI or a Purchasing Managers' Index (PMI) is an indicator of business activity — both in the **manufacturing and services sectors**. It is **calculated separately** for the manufacturing and services sectors and then a **composite index is constructed**.
- **Statement 2 is incorrect:** It is a survey-based measure that asks the respondents about changes in their **perception of some key business variables from the month (not year) before**. The PMI is derived from a series of **qualitative questions**. Executives from a reasonably big sample, running into hundreds of firms, are **asked whether key indicators such as output, new orders, business Expectations and employment** were stronger than the **month** before and are asked to rate them.
 - The PMI is usually released at the **start of the month**, much before most of the official data on industrial output, manufacturing and GDP growth becomes available. It is, therefore, considered a **good leading indicator of economic activity**

Extra Edge by ONLYIAS

• Additional Facts about PMI

- Purchasing Managers' Index data are compiled by IHS Markit for **more than 40 economies worldwide**.
- **What does a reading above/below 50 mean in PMI parlance ?**
 - Higher the difference from **50**, greater the **Expansion or contraction**. The rate of Expansion can also be judged by **comparing the PMI with that of the previous month data**. If the figure is **higher than the previous month's then the economy is Exp:anding at a faster rate**. If it is lower than the previous month then it is growing at a lower rate.
 - For example, recently, Data released by analytics firm IHS Markit showed purchasing managers' index (PMI) for **India's manufacturing activity in April 2021**, marginally rose to **55.5** after declining to a seven month low in **March 2021 at 55.4**. This has two inferences:
 1. First, There was an **overall increase** in the manufacturing activity in the month of **April** (since the **reading is above 50**)
 2. The **manufacturing activity in the month of April** was **slightly better** than that in the month

of March (since
55.5 > 55.4)

NEWS: [Factory orders, production rise at slowest rates in 8 months in April in India: PMI - The Hindu](#)

Q.78) Ans: B

Exp:

Maximum Admissible Export Quota:

- In 2019, *the Ministry of Consumer Affairs, Food & Public Distribution* notified a Scheme for evacuation of surplus stock of sugar from the country. Under this scheme, **Maximum Admissible Export Quantity (MAEQ)** is fixed among sugar mills for Export.
- The Govt. provides a lump sum Export subsidy towards Expenses on marketing costs including handling, upgrading and other processing costs and costs of international and internal transport and freight charges on Export of sugar.
- To encourage sugar industry to start Exploring possibility of Export of sugar, **the customs duty on Export of sugar has been withdrawn**

NEWS + SOURCE: [Sugar output rises by 41 lakh tonnes: ISMA - The Hindu](#)

Q.79) Ans: D

Exp:

- **Principle of Non-Refoulement**
 - Non-refoulement is a **fundamental principle of international law** that forbids a country **receiving asylum seekers from returning them to a country in which they would be in likely danger of persecution based on "race, religion, nationality, membership of a particular social group or political opinion"**
- Recently, the High Court of Manipur allowed **seven Myanmar nationals, who entered India secretly following the February military coup**, to travel to New Delhi to seek protection from

the **United Nations High Commissioner for Refugees (UNHCR)**.

- Manipur High Court said that **Article 21 (Right to life)** encompasses *the right of non-refoulement*.

NEWS : [Manipur HC allows 7 Myanmar refugees to approach UNHCR - The Hindu](#)

Q.80) Ans: C

Exp:

- **Options 1 is incorrect:**

- India has not signed the The 1951 Refugee Convention and its 1967 Protocol which define the **term 'refugee' and outline the rights of refugees, as well as the legal obligations of States to protect them.** UNHCR serves as the 'guardian' of the 1951 Convention and its 1967 Protocol.
- The line of argument is that **borders in South Asia are extremely porous and any conflict can result in a mass movement of people.** This can have two results
 - First, a strain on **local infrastructure and resources** in countries that are poorly equipped to deal with sudden spikes in population.
 - Second, it can **upset the demographic balance**, a tinderbox in South Asia.

- **Options 2 and 3 are correct**

- The **United Nations International Covenant of Civil and Political Rights (ICCPR)** attempts to ensure the protection of civil and political rights. It was adopted by the United Nations' General Assembly on December 19, 1966, and it came into force on March 23, 1976. The covenant commits its parties to respect the civil and political rights of individuals, including the right to life, freedom of religion, freedom of speech, freedom of assembly, electoral rights and rights to due process and a fair trial
 - **India** ratified the covenant in **1979**

- The **Universal Declaration of Human Rights (UDHR)** is an international document adopted by the **United Nations General Assembly** that enshrines the rights and freedoms of all human beings. The various articles detail an individual's "**basic rights and fundamental freedoms**" and affirming their universal character as inherent, inalienable, and applicable to all human beings
 - India was one of the **48 founding signatories of the UDHR in 1948**

NEWS : [Manipur HC allows 7 Myanmar refugees to approach UNHCR - The Hindu](#)

Q.81) Ans: C

Exp:

- In the ongoing **Virtual Summit** between **India** and the **United Kingdom** :
 - An ambitious '**Roadmap 2030**' was adopted to **elevate** bilateral ties to a '**Comprehensive Strategic Partnership**'. The Roadmap will pave the way for a **deeper and stronger engagement over the next ten years** in the key areas of people to people contacts, trade and economy, defence and security, climate action and health.
 - The two Prime Ministers launched an '**Enhanced Trade Partnership**' (ETP) to unleash the trade potential between the **5th and 6th largest economies** of the world and by setting an ambitious target of more than **doubling bilateral trade by 2030**. As part of the ETP, India and the UK agreed on a roadmap to **negotiate a comprehensive and balanced FTA**, including consideration of an **Interim Trade Agreement** for delivering early gains.
 - The UK is India's **second largest partner in research and innovation collaborations**. A new India-UK '**Global Innovation Partnership**' was announced at the Virtual Summit that aims to support the transfer of

inclusive Indian innovations to select developing countries, starting with **Africa**.

- India and the UK **launched a comprehensive partnership on migration and mobility** that will facilitate greater opportunities for the mobility of students and professionals between the two countries.

NEWS :

<https://pib.gov.in/PressReleasePage.aspx?PRID=1715968>

Question style

IAS //2019

Recently, India signed a deal known as 'Action Plan for Prioritization and Implementation of Cooperation Areas in the Nuclear Field' with which of the following countries?

- ☐ Japan
- ☐ Russia
- ☐ The United Kingdom
- ☐ The United States of America

IAS /Current Affairs/2017

'Broad-based Trade and Investment Agreement (BTIA)' is sometimes seen in the news in the context of negotiations held between India and

- ☐ European Union
- ☐ Gulf Cooperation Council
- ☐ Organization for Economic Cooperation and Development
- ☐ Shanghai Cooperation Organization

Q.82) Ans: C

Exp:

- **Statement 1 is correct:** Article **136** bestows the Supreme Court **discretion to entertain appeals in suitable situations, not otherwise** provided for in the Constitution. The SC may exercise this discretionary power to grant special leave to appeal any judgment or decree in any matter or case, made by any court or tribunal in the country.
 - The appeal can be made in a case where a **substantial question of law** is involved or where **gross injustice** has been observed (**Sanwat Singh v. the State of Rajasthan**)
 - A study of the SLPs, shows that the **SC grants leave only in case of**

exceptional situations and follows well-established judicial procedures in exercising the discretionary power

- **Statement 2 is correct:** The Special Leave Petition shall not apply to any judgment or order handed down by any court or tribunal involving the armed forces. This is the **only** exclusion as is given in the clause 2 of Article 136.

NEWS: 1. [Election Commission divided: One EC objects to panel asking court for gag order on media | India News, The Indian Express](#)

2. [The Election Commission needs to act to retrieve and restore its hard-earned credibility \(indianexpress.com\)](#)

Q.83) Ans: A

Exp:

- **How Commercial Oxygen Concentrators work ?**
 - With the demand for medical oxygen continuing unabated and several States struggling to keep pace with demand, the oxygen concentrator has emerged as a sought after device.
 - An oxygen concentrator takes in air and separates the oxygen and delivers it into a person via a nasal cannula. Air is 79% nitrogen and 21% oxygen and a concentrator that works by plugging into a source of electricity delivers air that is upto 95% oxygen. In respiratory infections that causes oxygen saturation levels to dip below 90%, having an external device supply pure oxygen eases the burden on the lungs.
 - A concentrator consists of a compressor and sieve bed filter. The former squeezes atmospheric air and also adjusts the pressure at which is delivered. The sieve bed is made of a material called

ZEOLITE that separates the nitrogen.

- There are two sieve beds that work to both release oxygen into a tank that's connected to the cannula as well as release the separated nitrogen and form a continuous loop that keeps producing fresh oxygen.

NEWS: [Explainer | How does an oxygen concentrator help? - The Hindu](#)

Questions style: Similar to this framework, *where the process is explained to the candidate* and only a **key factual thing is asked**.

IAS //2020

The experiment will employ a trio of spacecraft flying in formation in the shape of an equilateral triangle that has sides one million kilometres long, with lasers shining between the craft." The experiment in question refers to

- ☐ Voyager-2
- ☐ New Horizons
- ☐ LISA Pathfinder
- ☐ Evolved LISA

Q.84) Ans: D

Exp:

- **All the given options are grounds for Contempt of Court and hence none is incorrect:**
 - The Central and state governments have **recently received flak** from the higher judiciary of the country for the way it has gone about its business in handling the covid-19 crisis. The **Delhi High Court** even warned Centre that it will face **contempt charges** if supplies of essential medical equipment were not ramped up.
 - According to the **Contempt of Courts Act, 1971**, contempt of court can either be **civil contempt** or **criminal contempt**.
 - **Civil contempt** means **wilful disobedience** of any

judgment, decree, direction, order, writ or other process of a court, or wilful breach of an undertaking given to a court.

- **Criminal** contempt, on the other hand, is attracted by the **publication** (whether by words, spoken or written, or by signs, or by visible representations, or otherwise) of any matter or the doing of any other act whatsoever which:

- scandalises or **tends to scandalise**, or lowers or tends to lower the authority of, any court;
- prejudices, or interferes or **tends to interfere** with, the due course of any **judicial proceeding**;
- **interferes** or tends to interfere with, or obstructs or tends to obstruct, the **administration of justice** in any other manner.

- Also, Articles **129** and Article **215** Of the Indian Constitution talk about the **inherent contempt powers** of the **Supreme Court** and **High Courts** respectively.

NEWS: [HC warns Centre of contempt case if Delhi doesn't get oxygen, state says can't make it ego issue - Cities News \(indiatoday.in\)](#)

Q.85) Ans: C

Exp:

- The Wild Life (Protection) Act, 1972 is an Act of the Parliament of India enacted for protection of plants and animal species. It has **six** schedules which give varying degrees of protection. **Schedule I and part II of Schedule II provide absolute protection** - offences

under **these** are prescribed the highest penalties.

- The **Indian pangolin, Mouse Deer, Chinese Pangolin, Slender Loris and Himalayn Tahr** have been included under the Schedule 1 of the WPA, 1972.
- Species listed in **Schedule III and Schedule IV** are also protected, but the **penalties** are much **lower**. Animals under Schedule V, e.g. **common crows, fruit bats, rats and mice**, are legally considered **vermin** and may be **hunted freely**.

NEWS: [Wildlife forensics helps cause of pangolins - The Hindu](#)

Q.86) Ans: C

Exp:

- **Statement 1 is incorrect:** The Constitution **does not have any provision** that **explicitly bars** a **resident** of a **particular** state from becoming the governor of the state. It is a **convention that has developed** over time that the governor of a state is generally from from another state
 - The only qualifications for becoming a governor according to **article 157** are:
 - Governor should be a **citizen** of India
 - The Governor must have completed the age of **thirty-five years**
- **Statement 2 is incorrect:**
 - According to **article 174** of the constitution, the Governor shall from time to time summon the House or each House of the Legislature of the State to meet **at such time** and place as **he thinks fit**.
 - In normal circumstances, the power vested with the Governor **under Article 174 to summon, prorogue and dissolve** the house(s) must be **exercised in consonance with the aid and advice of the chief minister** and his **council of ministers**. In such a situation, he is **precluded [from taking]**

an individual call on the issue at his own will, or in his own discretion.

- However, When the CM appears to have lost the majority and the legislative members of the House propose a no-confidence motion against the CM, then the Governor can decide on his or her own (discretionary power) on summoning the House. But the actions of the Governor, when using his discretionary powers, can be challenged in court.
- **Statement 3 is correct:** A number of rulings by the Supreme Court have settled the position that the Governor cannot refuse the request of a Cabinet that enjoys the majority in the House unless it is patently unconstitutional (Read *Nabam Rebia (2016)*).
 - Even the Sarkaria Commission of 1983, which reviewed the arrangements between the Centre and the states, had said that “so long as the Council of Ministers enjoys the confidence of the Assembly, its advice in these matters, unless patently unconstitutional, must be deemed as binding on the Governor. It is only where such advice, if acted upon, would lead to an infringement of a constitutional provision, or where the Council of Ministers has ceased to enjoy the confidence of the Assembly, that the question arises whether the Governor may act in the exercise of his discretion.
- **Statement 4 is incorrect:** The Constitution does not lay down any grounds upon which a governor may be removed by the President. Generally, it is the union government which appoints and removes governors based on its own discretion.

NEWS: [Out of line: On West Bengal Governor's visit to violence-hit areas - The Hindu](#)

Q.87) Ans: B

Exp:

- According to a publication by the Zoological Survey of India, titled *Faunal Diversity of Biogeographic Zones: Islands of India* has for the first time come up with a database of all faunal species found on the island, putting the number at **11,009**. The documentation proves that the islands, comprising only **0.25%** of India's geographical area, are home to more than **10%** of the country's fauna species. Some of the endemic species are
 - Andaman drongo,
 - Andaman serpent-eagle
 - Hume's boobook
 - Andaman shama
 - Great nicobar Crake
 - Andaman nightjar
 - Narcondam hornbill

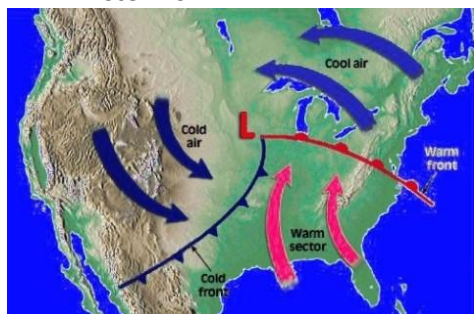
NEWS: [Green panel allows Great Nicobar plan to advance - The Hindu](#)

Q.88) Ans: A

Exp:

- **Statement 1 is incorrect:**
 - Tornado, a small-diameter column of violently rotating air developed within a convective cloud and in contact with the ground. Tornadoes occur most often in association with thunderstorms during the spring and summer in the mid-latitudes (between 20° and 60° N and S) of both the Northern and Southern Hemispheres.
 - Tornadoes have been reported on all continents except Antarctica. A tornado occurrence in Antarctica isn't impossible. For tornadoes to form, there needs to be a moist, warm climate, which simply isn't the case in Antarctica
- **Statement 2 is correct:**
 - Tornadoes have diameters on the scale of 100s of meters and are produced from a single convective storm (i.e. a thunderstorm or

cumulonimbus). A tropical cyclone, however, has a diameter **on the scale of 100s of kilometers** and consists of several to **dozens of convective storms**.



• **Statement 3 is incorrect:**

- While tropical cyclones form only on sea with temperatures more than 26-27 degrees Celsius and dissipate on reaching the land, on the other hand, Temperate cyclones or **ExtraTropical cyclones** can be **formed on both land and sea**.

- The origin and development of temperate cyclones is **made possible due to the formation of Polar Fronts over land** (shown in the figure) - that is **when polar and tropical air masses meet and form fronts in the mid latitudes**, especially in the **winter season**. The **extreme contrast in temperatures** along with **Rossby waves** and **jet streams** create conducive conditions for their origin.

NEWS: [300 injured in tornado strikes in two Chinese cities](#)

Q.89) Ans: C

Exp:

• **Statement 1 is incorrect:**

- Central Adoption Resource Authority (CARA) is a **statutory** body of the Ministry of Women & Child Development, Government of India. It functions as the **nodal body for adoption of Indian children** and is mandated to monitor and regulate **in-country and inter-country adoptions**.

- CARA is designated as the **Central Authority to deal with inter-country adoptions** in accordance with the provisions of the **Hague Convention on Inter-country Adoption, 1993**, ratified by Government of India in 2003

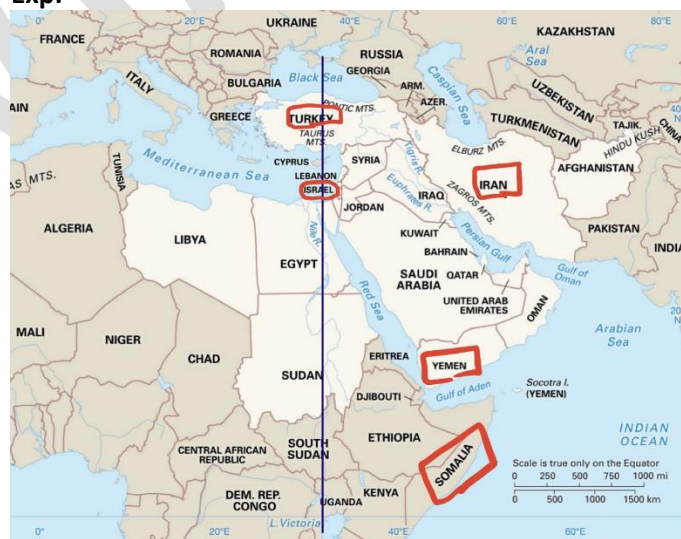
• **Statement 2 is incorrect:**

- The child could be an **"orphan, abandoned and surrendered child, declared legally free for adoption by the Child Welfare Committee"**, or a **"child or children of spouse from earlier marriage, surrendered by the biological parent(s) for adoption by the step-parent"**. One can also adopt a **child of a relative as defined under sub-section 52 of Section 2 of the Juvenile Justice Act, 2015**.

News: [Child Adoption in India \(indianexpress.com\)](http://indianexpress.com)

Q.90) Ans: A

Exp:



- The governance of the **Gaza Strip** is carried out by the **Hamas administration** which was recently involved in a **11-day bombing spree** with the **Israeli forces**.
- The Gaza Strip is a narrow piece of land towards the **mediterranean side of the Sinai peninsula**.
- Students are advised to be thorough with the map of the Middle East region.**

Q.91) Ans: B

Exp:

- Options 1 and 3 are correct
 - A default means **non-payment of a loan** availed by a borrower. A **willful defaulter is an entity** or a person that **has not paid the loan back despite the ability to repay** it.
 - A 'wilful default' would be deemed to have occurred if The Unit has defaulted in meeting its payment / repayment obligations to the lender
 - even when **it has the capacity to repay**.
 - and has not utilised the finance from the lender for the specific purposes for which finance was availed of but has **diverted the funds for other purposes**.
 - and has **siphoned off the funds** so that the funds have not been utilised for the specific purpose for which finance was availed of, **nor are the funds available with the unit in the form of other assets**.
 - and has also disposed off or **removed the movable fixed assets or immovable property given for the purpose of securing a term loan without the knowledge of the bank / lender**.

NEWS: [Wilful defaults took a turn for the worse in Apr-Dec amid pandemic \(livemint.com\)](https://www.livemint.com/Apr-Dec amid pandemic)

Q.92) Ans: D

Exp:

- Statement 1 is correct:
 - The **Montagu-Chelmsford reforms** led to the formation of the Council of State at the national level in 1919. Then the **Government of India Act of 1935** set up **bicameral legislatures** in

Indian provinces. It was under this law that a **Legislative Council first started functioning in Bengal in 1937**.

- Statement 2 is correct:
 - According to **Article 171**, The **size of the State Legislative Council** cannot be more than **one third of the membership of the State Legislative Assembly**. However, its size cannot be **less than 40 members**. These members elect the Chairman and Deputy Chairman of the State Legislative Council.
- Statement 3 is incorrect:
 - The setting up of a second chamber is not exclusively in the hands of the state legislative Assembly. The central government also has to pilot a Bill in Parliament.
 - **Article 169 of the Constitution** empowers the **Legislative Assembly** to create or abolish a Legislative Council by passing a resolution. The resolution has to be passed by two-thirds of the Assembly members. **Then a Bill to this effect has to be passed by Parliament (simple majority)**.
- Statement 4 is correct:
 - Under **Article 80**, the President can nominate persons having special knowledge or practical Experience **Literature, science, art and social service** to the Rajya Sabha
 - Under **Article 171**, The members to be **nominated by the Governor to the Legislative Council**, shall consist of persons having **special knowledge or practical Experience in Literature, science, art, co-operative movement and social service**.

NEWS: [West Bengal wants Upper House back: how states have Councils | explained News, The Indian express](https://www.thehindu.com/news/national/West-Bengal-wants-Upper-House-back-how-states-have-Councils-explained-news-the-indian-express)

Q.93) Ans: C

Exp:

• **Statement 1 is correct and statement 3 is incorrect:**

- The power to create new districts or alter or abolish existing districts rests with the State governments. **The Centre has no role to play in the alteration of districts or creation of new ones. States are free to decide. The Home Ministry comes into the picture when a State wants to change the name of a district or a railway station.** The State government's request is sent to other departments and agencies such as the Ministry of Earth Sciences, Intelligence Bureau, Department of Posts, Geographical Survey of India Sciences and the Railway Ministry seeking clearance

• **Statement 2 is correct**

- This can either be done **through an executive order or by passing a law in the State Assembly.** Many States prefer the executive route by simply issuing a notification in the official gazette. Many States prefer the executive route by simply issuing a notification in the official gazette. States argue that smaller districts lead to better administration and governance.

• Note that the **Global Economic Prospects (GEP) report** is published by the **World Bank**

Q.95) Ans: A

Exp:

- **Statement 1 is correct:** The Central Bureau of Investigation (CBI) was set up in **1963** by an executive resolution of the Ministry of Home Affairs. Later, **it was transferred to the Ministry of Personnel, Public Grievances and Pensions** and now it enjoys the status of an attached office

- The CBI is the **main investigating agency** of the Central Government. It also provides **assistance to the Central Vigilance Commission and Lokpal.**
- CBI is India's officially designated single point of **contact for liaison with the Interpol.**

• **Statement 2 is incorrect:**

- Before the Lokpal Act was legislated, **the CBI director was appointed by the Delhi Special Police Establishment Act, 1946.** Now, **the Lokpal Act, 2013** governs the appointment of the CBI director.
- The CBI Director is appointed, **for not less than a term of 2 years**, by the Appointment Committee on recommendation of the Selection Committee as mentioned in **DSPE Act 1946** amended through the **Lokpal & Lokayukta Act, 2013.**

- The **Appointment Committee** consists of:

- **Prime Minister** – Chairperson
- **Leader of Opposition of Lok Sabha** or the Leader of the single largest opposition party in the Lok Sabha, if the former is not present due to lack of mandated strength in the Lok Sabha - member
- **Chief Justice of India** or a Supreme Court Judge recommended

NEWS: [explained: The why and how of creating a new district - The Hindu](#)

Q.94) Ans: B

Exp:

- The **World Economic Situation and Prospects 2021**, produced by the **United Nations Department of Economic and Social Affairs (UN DESA)**, said the world economy was hit by a **once-in-a-century crisis, a Great Disruption** unleashed by the Covid-19 pandemic in 2020.
 - India's **economy is projected to grow at 7.3 per cent in 2021**, even as it is estimated to contract by **9.6 per cent in 2020** as **lockdowns** and other efforts to control the Covid-19 pandemic slashed domestic consumption.

by the Chief Justice – member.

- **Statement 3 is incorrect:**
 - The CBI **remains an executive body/ attached office** and has no statutory backing as of now. It derives its powers from the **Delhi Special Police Establishment Act, 1946**.

Q.96) Ans: D

Exp:

- **Assam Rifles** is one of the **six central armed police forces (CAPFs)** under the **administrative control** of the **Ministry of Home Affairs (MHA)**. The other forces being the Central Reserve Police Force (**CRPF**), the Border Security Force (**BSF**), the Indo-Tibetan Border Police (**ITBP**), the Central Industrial Security Force (**CISF**) and the Sashastra Seema Bal (**SSB**)
- However, **It is the only paramilitary force** with a **dual control structure**. While the administrative control of the force is with the MHA, **its operational control is with the Indian Army, which is under the Ministry of Defence**.
 - This means that **salaries and infrastructure for the force is provided by the MHA**, but the **deployment, posting, transfer and deputation of the personnel is decided by the Army**. All its senior ranks, from DG to IG and sector headquarters, are manned by officers from the Army. The force is commanded by Lieutenant General from the Indian Army.
 - The force is the **only central paramilitary force (CPMF)** in **real sense** as its operational duties and regimentation are on the lines of the Indian Army. However, its recruitment, perks, promotion of its personnel and retirement policies are governed according to the rules framed by the MHA for CAPFs.

NEWS: [Overstepping the norm - The Hindu](#)
[Who controls Assam Rifles?](#)
indianexpress.com

Q.97) Ans: A

Exp:

- **Statement 1 is correct and Statement 2 is incorrect:**
 - **Corticosteroids, commonly called steroids**, are **man-made drugs** that **closely resemble cortisol**, a hormone that the adrenal glands produce naturally.
 - Steroids work by **decreasing inflammation** and **reducing** the activity of the immune system. **Inflammation** is a process in which the **body's white blood cells and chemicals can protect against infection and foreign substances** such as bacteria and viruses.
 - In certain diseases, however, the body's defense system (immune system) doesn't function properly. Steroids reduce the production of chemicals that cause inflammation while also **reducing the activity of the immune system by affecting the way white blood cells work**.
- **Statement 3 is incorrect:**
 - **Not all patients will develop side effects**. How often any side effect occurs **varies from person to person**. If steroid use is brief (from a few days to a few weeks), it is possible that **none of the common side effects will occur**. However, if steroid use involves high doses and is prolonged (for a few months to several years), an increase in the number of side effects might occur. **The prolonged use of high dose steroids is justified only for severe illnesses that represent serious risks to the patient**.

SOURCE : [Corticosteroids \(clevelandclinic.org\)](http://clevelandclinic.org)

Q.98) Ans: B

Exp:

• Liquidity Risk:

- Liquidity risk involves the risk of not being able to sell an instrument or close a position when required without facing significant costs. The reserves need to have a high level of liquidity at all times in order to be able to meet any unforeseen and emergency needs.
- Any adverse development on the external front would pose a demand on our forex reserves and, hence, the investment strategy needs a highly liquid portfolio. The choice of instruments determines the liquidity of the portfolio. For example, in some markets, treasury securities could be liquidated in large volumes without much distortion of the price in the market and, thus, can be considered as liquid.
- Except fixed deposits with the BIS/ commercial banks overseas / central banks and securities issued by supranationals, almost all other types of investments are highly liquid instruments which could be converted into cash at short notice. The Reserve Bank closely monitors the portion of the reserves, which could be converted into cash at a very short notice, to meet any unforeseen/ emergency needs.

NEWS: [The Hindu explains | What has the RBI cautioned against in its latest Financial Stability Report?](#)

[Uncertainty, risks cloud outlook: RBI - The Hindu](#)

Q.99) Ans: C

Exp:

- The forum is an abbreviation for **Cooperation between China and Central and Eastern European Countries**. It is an initiative by the **Chinese Ministry of Foreign Affairs** to promote business and investment relations between China and 16 countries of CEE (CEEC).
 - The **format was founded in 2012** in **Warsaw** to push for the cooperation of

the “17+1” (the 17 CEE countries and China).

- Its goals are to promote the **Chinese Belt and Road Initiative** and enhance **cooperation** in the fields of infrastructure, transportation, and logistics, trade and investment

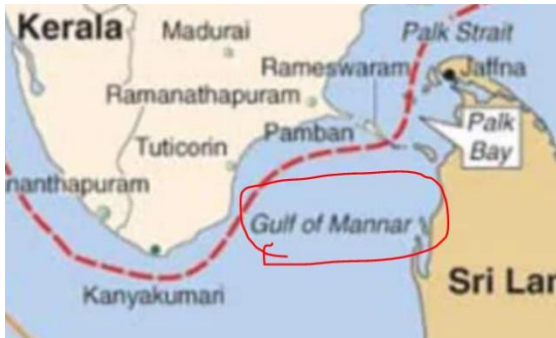
NEWS: [Lithuania quits ‘divisive’ China group - The Hindu](#)

Q.100) Ans: B

Exp:



- **Statement 1 is incorrect:** The Tungabhadra River is formed by the confluence of the **Tunga River and the Bhadra River at Koodli**, which is **towards the north** of the the source of River Cauvery at **Talakaveri in the Brahmagiri range** in the Western Ghats.
- **Statement 2 is correct:** There are many tributaries of River Cauvery, including Harangi, **Hemavati, Kabini, Bhavani**, Lakshmana Tirtha, Noyyal and Arkavati.
- **Statement 3 is incorrect:** The river mouth lies between the Thanjavur and the Tiruchirappalli districts **where it empties into the Bay of Bengal**.



- The **Gulf of Mannar** is located further south towards the tip of **India's peninsula**.

NEWS: [Oppose Mekedatu project, urge farmers - The Hindu](#)